

# READING IS NOT WRITING: SEMANTIC COMPETITION FOR PROBE DIRECTIONS IN MEDICAL VISION-LANGUAGE MODELS

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## ABSTRACT

A probe direction can reproducibly change a clinical answer without earning attribution to its training label. Random and sham controls test whether an intervention exceeds generic perturbations, but leave its advantage over other clinical directions untested. We study this distinction through semantic competition at fixed consumed visual loci in LLaVA-1.5-7B and Qwen2.5-VL-7B on NIH ChestX-ray14, holding direction construction, dose, patients, and endpoint constant within each comparison. Two Qwen findings separate intervention efficacy from label-specific attribution. First, locked-dose Effusion steering replicates on 400 independent patients beyond random p95 and sham, yet its effect is smaller than the strongest fixed clinical competitor’s by 0.0573 (ownership contrast  $-0.0573$ , 95% CI  $[-0.0694, -0.0450]$ ). Second, Mass outperforms all five clinical competitors on 200 independent patients under both *show* A/B mappings, with positive simultaneous lower bounds, while both *is* margins are negative. Paired wording analyses descriptively support this dependence. Larger effects among 119 random directions defeat both registered Mass specificity criteria. Diagnostic benefit is also unsupported: all confirmation AUROC-change intervals cross zero, and Brier error increases in every prompt condition. Semantic competition measures relative advantage within a fixed clinical family. These findings establish that successful steering can coexist with a clinical attribution deficit, and that clinical advantage can depend on wording.

## 1 INTRODUCTION

A successful intervention poses an attribution question: does the direction’s training label explain its privileged effect on the answer? Linear probes identify information available to restricted readouts (Alain & Bengio, 2017; Hewitt & Liang, 2019), and decodability can persist without task relevance (Ravichander et al., 2021). Activation steering goes further by changing a representation and measuring the downstream response (Turner et al., 2023; Panickssery et al., 2024). Even after this intervention succeeds, assigning its effect to the probe’s clinical label requires a comparison with semantic alternatives.

Random directions test generic sensitivity at a matched dose, while a coordinate-permutation sham tests dependence on the direction’s coordinate arrangement. Neither asks whether an equally constructed direction bearing another clinical label produces a larger response to the same question. Separability-trained concept vectors can absorb distractor signals (Pahde et al., 2025). Semantic competition tests the resulting attribution directly by comparing labeled writers after a controlled write effect has been established.

We study NIH ChestX-ray14 with fixed direction bundles at the consumed final visual blocks of LLaVA-1.5-7B and Qwen2.5-VL-7B. For fixed yes/no prompts,  $W_{q,d}$  is the paired mean change in question  $q$ ’s yes probability under direction  $d$ . Ownership  $O_q = W_{q,q} - \max_{d \neq q} W_{q,d}$  compares the named direction with its strongest clinical competitor (Figure 1). Contenders share construction, locus, dose, patients, and endpoint. Ownership denotes relative advantage within this fixed clinical family, not a necessary condition for concept participation in computation. Finding-present logit margins align the endpoint across answer mappings.

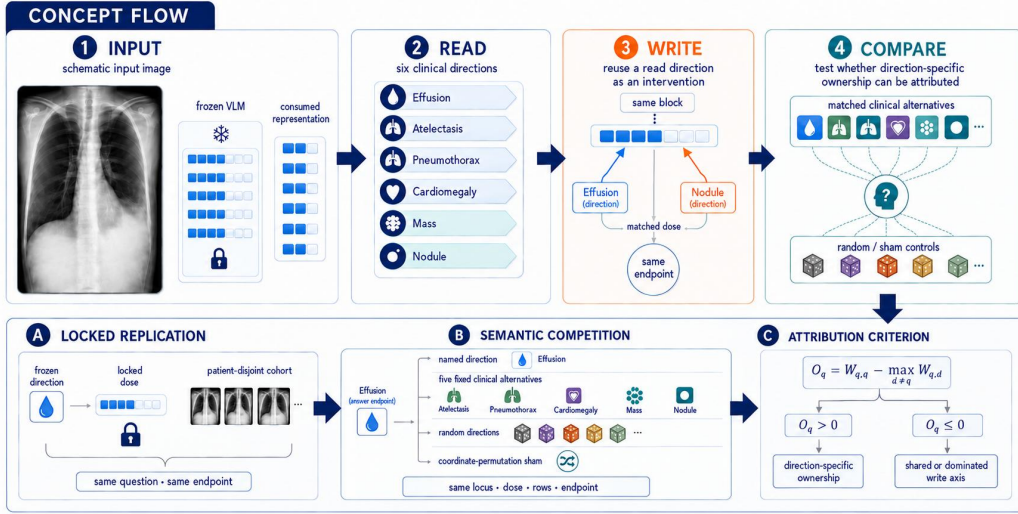


Figure 1: **Semantic competition under a fixed prompt.** At the consumed visual block, Effusion and Nodule directions both increase the Effusion answer in the patient-disjoint Qwen result, but Nodule is stronger.

Two findings organize the results. Qwen Effusion steering replicates at a locked dose on 400 independent patients beyond random p95 and sham, yet loses to the strongest clinical competitor by 0.0573 (95% CI for  $O_{\text{Effusion}}$ :  $[-0.0694, -0.0450]$ ; Sections 4.1–4.2). Thus successful intervention coexists with a clinical attribution deficit.

Mass’s clinical advantage reproduces on 200 independent patients under both *show* A/B mappings, with positive simultaneous lower bounds against all five competitors, while both *is* margins are negative (Section 4.3). Paired wording analyses are descriptive. Larger effects among 119 random directions defeat both registered specificity criteria. All confirmation AUROC-change intervals cross zero and Brier error increases, leaving diagnostic benefit unsupported.

The remaining experiments delimit these attribution findings through boundary analyses of broader competition, mechanism, LLaVA behavior, and natural paired response (Section 4.4).

## 2 RELATED WORK

**Controlled probing.** Linear classifiers have long been used to track information across network depth (Alain & Bengio, 2017). A probe score reflects both the representation and the learner fitted on top of it, motivating matched-capacity probes and control tasks (Hewitt & Liang, 2019). The broader probing literature distinguishes information that a readout can recover from information that a model uses for its task (Belinkov, 2022). On synthetic tasks, high probing accuracy can persist for properties that the model provably does not need, directly separating decodability from task relevance (Ravichander et al., 2021). Causal erasure methods pursue the complementary test: remove linearly identified information and measure the behavioral consequence (Ravfogel et al., 2020; Elazar et al., 2021). Our protocol brings these concerns to medical images with patient-disjoint splits, a common 512-dimensional projection, repeated recurring-type controls, and bootstrap intervals. It asks about one verified consumed visual locus rather than selecting the best layer after observing test results.

**Representation interventions.** Activation engineering changes intermediate representations at inference time and measures how outputs respond (Turner et al., 2023; Panickssery et al., 2024). A probe normal is an appealing intervention direction because it supplies a semantic label and a signed vector from the same representation. Separability-trained concept vectors can, however, absorb distractor signals and diverge from the concept direction they are meant to model (Pahde et al., 2025). We ask a distinct operational question: when that labeled normal is inserted into a model’s forward path, does it outperform equally fitted semantic alternatives? Random vectors and a coordinate-

permutation sham address generic magnitude and construction effects; fixed clinical normals address attribution. Our statistic recomputes the maximum alternative clinical effect inside each patient bootstrap, making direction specificity one familywise endpoint rather than five separate pairwise claims.

Intervention families answer different causal questions. Iterative nullspace projection and amnesic probing test whether removing linearly recoverable information damages a downstream task (Ravfogel et al., 2020; Elazar et al., 2021). Additive steering instead tests whether a chosen direction can write a change into the forward pass (Turner et al., 2023). The second test needs an operating scale: an unnormalized vector can appear effective simply because its norm is large, while a fixed Euclidean step can represent very different perturbations across layers or architectures. We use each token’s own activation norm as the scale and compare every direction at the same  $\alpha$ . This choice makes the random, sham, target, and clinical interventions commensurate within a registered cell. It also keeps a generic sensitivity result separate from a claim that the probe normal uniquely represents its named clinical concept.

**Medical and multimodal VLM interpretation.** Depth-wise probing of medical multimodal models has identified losses and gains of classification information across vision towers, connectors, language layers, and semantic outputs (Zhu et al., 2026). Layer-wise probes of MedCLIP also expose acquisition and device shortcuts in chest radiographs (Pedersen et al., 2026). These studies motivate tight nuisance and data controls, but they do not intervene on the decoded clinical direction at the same locus. Conversely, controlled visual-concept experiments in lightweight driving VLMs combine probing with probe-vector steering and identify cases where information is linearly present but generation still fails (Theodoridis et al., 2026). Targeted edits in general VLMs likewise connect linearly available visual features to downstream predictions (Rajaram et al., 2025). We focus on 7B medical-question answering, preserve the clinical patient split, and compare the target probe normal with equal-norm random, sham, and clinical directions.

Together, this literature leaves a specific identification gap. A probe is fitted as a reader, while steering reuses its normal as a writer; successful writing does not establish that the read label specifically names the effective direction. Unlike tests that only add, remove, or trace the target direction, semantic competition estimates its advantage over equally constructed named alternatives and treats the maximum alternative as one familywise endpoint. Our contribution is a matched, in-forward-path attribution test that aligns controlled reading, non-semantic intervention controls, semantic competitors, and familywise uncertainty at one consumed representation.

### 3 EXPERIMENTAL DESIGN

**Data and split.** We use NIH ChestX-ray14 (Wang et al., 2017), whose findings are mined from radiology reports. A deterministic hash of patient identity assigns rows to disjoint train, validation, and test partitions. The registered manifest contains 18,212 training images and 5,343 test images from 1,659 test patients. The three cells pair LLaVA-1.5-7B with Effusion and Edema, and Qwen2.5-VL-7B with Effusion. All reported uncertainty clusters by patient. Prompts and exact run identifiers appear in Appendix A.

**Models and consumed loci.** For LLaVA-1.5-7B (Liu et al., 2024) and Qwen2.5-VL-7B (Bai et al., 2025), we use the final visual block consumed by the connector or merger. Writing  $\alpha$  for the signed intervention dose, synthetic hook tests establish downstream-path membership, a bitwise  $\alpha = 0$  no-op, and changed downstream representations and logits at nonzero dose. Probe and intervention therefore share one consumed representation; exact modules appear in Appendix A.

**Controlled decodability.** For image  $i$ , let  $\mathbf{h}_{it} \in \mathbb{R}^D$  denote the  $D$ -dimensional representation of visual token  $t$  at the consumed final block; mean pooling over tokens yields  $\mathbf{h}_i$ . A fixed seed-0 Gaussian matrix  $R \in \mathbb{R}^{D \times 512}$  projects every model to a common readout dimension; featurewise scale  $s$  is fitted on training rows only. We then fit logistic regression with  $C = 1$ , maximum 2,000 iterations, and seed 0. Each of 20 control seeds maps the recurring input type (AP view indicator, sex indicator, and age decade, giving 39 strata) to a random label. Let  $y$  be the real test labels and  $\hat{y}$  the fitted probe’s continuous scores. For control seed  $k \in \{0, \dots, 19\}$ ,  $\tilde{y}_k$  and  $\hat{y}_k$  are the random

labels and corresponding control probe scores. Selectivity compares their AUROCs:

$$S = \text{AUROC}(y, \hat{y}) - \frac{1}{20} \sum_{k=0}^{19} \text{AUROC}(\tilde{y}_k, \hat{y}_k). \quad (1)$$

We report the real AUROC, control distribution, and  $S$  with 2,000 patient-cluster bootstrap resamples. A positive interval for  $S$  means that the target is more readable than same-capacity nuisance-type label maps. It does not yet imply downstream use. Paired contrasts reuse identical ordered test rows and bootstrap draws.

**Probe-normal intervention.** Let  $\beta_c$  be the projected-space logistic coefficient for the positive class of concept  $c$ . We lift it to the consumed  $D$ -dimensional representation and normalize it exactly as

$$\hat{\mathbf{w}}_c = \frac{R(\beta_c \oslash \max(s, 10^{-8}))}{\|R(\beta_c \oslash \max(s, 10^{-8}))\|_2}. \quad (2)$$

The positive-class coefficient fixes the direction’s clinical pole. At every visual token  $t$ , the registered intervention is

$$\mathbf{h}'_{it} = \mathbf{h}_{it} + \alpha \|\mathbf{h}_{it}\|_2 \hat{\mathbf{w}}_c. \quad (3)$$

Here the sign of  $\alpha$  selects the positive or negative clinical pole, while  $|\alpha|$  is the perturbation norm as a fraction of each token’s activation norm. At the first answer position,  $P(\text{yes}) = \sigma(\ell_{\text{yes}} - \ell_{\text{no}})$ , where each log probability is the maximum over case and leading-space variants that tokenize to one token; the endpoint is its paired mean change from the common unsteered baseline. Original cells use concept doses  $\{-1, -0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5, 1\}$  on 200 held-out test images. Controls are evaluated at the six nonzero doses excluding  $\pm 0.1$ : 20 isotropic random unit directions, one sham formed by a fixed coordinate permutation of  $\hat{\mathbf{w}}_c$  and renormalization, and fixed alternative clinical probe normals. Every control receives the same signed dose on the same rows. Dose monotonicity over  $[-0.5, 0.5]$  is descriptive.

**Original-cell decision rule.** For each controlled nonzero dose, the concept-consistent change is  $\text{sign}(\alpha)\Delta(\alpha)$ : positive steering should increase  $P(\text{yes})$ , and negative steering should decrease it. The primary effect is the maximum concept-consistent change over those doses. A cell clears the generic intervention gate only if this effect leaves the same-dose random 5th–95th percentile interval in the concept-consistent direction and exceeds the maximum absolute sham effect. The random p95 is a registered reference gate over 20 directions, not a conventional 5% hypothesis test. The conjunction is fixed across the three cells.

**Direction-specificity supplement.** The initial experiment prespecified the random and sham comparisons; its clinical-direction analysis was descriptive. The Nodule response motivated a separately registered supplement at the locked Qwen dose,  $\alpha = +0.25$ . We select one row by label-blind hash for each of 400 ChestX-ray14 test patients absent from the original 200-image intervention cohort. The semantic family contains all five non-target normals from the pre-existing Qwen direction bundle (Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule), rather than directions chosen from the supplement outcome. Each uses the same training rows, projection, scaling, classifier, lifting rule, and normalization as Effusion. Together with a common baseline, 20 random directions, and the sham, the grid produces 28 configurations and 11,200 per-image outcomes; exact row selection appears in Appendix A.

Semantic ownership is a contrast on a direction–question response surface, not a property of a direction in isolation. Let  $W_{q,d}$  denote the paired mean change in question  $q$ ’s yes probability after intervening with direction  $d$  at the locked dose. A large  $W_{q,d}$  establishes that  $d$  can write  $q$ ’s answer; it does not establish that  $d$  owns that answer. Within a fixed clinical family, the ownership contrast is

$$O_q = W_{q,q} - \max_{d \neq q} W_{q,d}. \quad (4)$$

The registered supplement estimates the Effusion row of this surface, so the reported ownership contrast  $O_{\text{Effusion}}$  is Effusion minus its maximum alternative. The maximum is recomputed inside each of 5,000 patient-bootstrap draws. Thus  $O_{\text{Effusion}} > 0$  means Effusion beats every member of the fixed clinical family, while  $O_{\text{Effusion}} < 0$  means at least one alternative is stronger. This operational ownership criterion measures relative advantage, not a necessary condition for a concept to participate

in computation. A negative contrast is compatible with correlated findings, mixed representations, or shared channels; the comparison does not distinguish those explanations. Direction specificity requires a positive Effusion effect above random p95 and absolute sham, together with a one-sided 95% lower bound for  $O_{\text{Effusion}}$  above zero.

**Causal-ownership matrix.** To test whether another clinical normal better matches the answer pathway, we prospectively cross six Qwen directions (Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule) with their six matching questions on 400 new patients. Every question uses the same  $\alpha = +0.25$ , 119 random directions, and a sham. For question  $q$ , ownership requires the diagonal effect  $W_{q,q}$  to exceed every other clinical direction, every random direction, and the absolute sham effect. A centered studentized max- $T$  bootstrap gives simultaneous 95% lower bounds over all 30 clinical contrasts. The shared-alias family contains two statistics per direction: mean excess over the matching diagonals on its five off-diagonal questions and signed mean effect on those questions. These give 12 max- $T$  bounds; the effect must also exceed the random alias maximum and maximum absolute sham effect, detailed in Appendix C.

**Within-patient input closure.** Because the matrix confirms neither ownership nor an alias, the registered fallback tests Consolidation on 50 fresh patients who each have one hash-selected positive and negative study. Let  $\gamma_i$  be the positive-minus-negative pooled `vis.last` displacement along the train-fitted Consolidation normal. The write dose is  $\alpha_i = \text{clip}(\gamma_i / \|\mathbf{h}_{it}\|, -0.25, 0.25)$  on the negative study. The closure gain is the reduction in its absolute  $P(\text{yes})$  gap from the paired positive study. Confirmation requires a positive displacement lower bound, a concept gain above all 119 random directions and the sham, and a positive familywise lower bound against Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule over 5,000 paired bootstraps.

**Answer-encoding crossover.** We next register a paired sensitivity study on 200 patients from the causal-ownership matrix cohort; the other 200 form a patient-disjoint capability-calibration set. This is not the earlier Effusion specificity cohort. Six questions cross the original yes/no prompt with A-present and B-present prompts, which exchange the finding-to-letter mapping. Each uses both  $\alpha = \pm 0.25$ , six clinical directions, 20 random directions, and a sham. A question enters the primary endpoint only if both clean coded mappings have AUROC one-sided 95% lower bounds above 0.5 and at least ten patients per class on calibration. At  $\alpha = +0.25$ , we compare squared mean effects in components that retain or reverse raw A-minus-B sign under the mapping exchange, using an unbiased estimator and 2,000 joint patient-bootstrap draws. Appendix H defines the components and full control rule. Descriptive clinical competition uses the contrast in Equation 4, replacing probability change with finding-present logit-margin change within each prompt.

**Independent Mass confirmation.** We prospectively test Mass on 200 patients absent from all preceding Qwen intervention cohorts, retaining the crossover’s 200 calibration patients. Two question wordings (*show*, *is*) cross explicit yes/no, A-present, and B-present instructions; the original yes/no prompt is a seventh anchor. At  $\alpha = +0.25$ , each cell receives six clinical directions, 119 random directions, and a sham. The four coded cells share 20 clinical comparisons with simultaneous 95% lower bounds. Confirmation requires positive bounds and Mass effects above every random effect and absolute sham, either across all four coded cells or both original-wording cells. Appendix I specifies calibration and paired descriptive contrasts.

## 4 RESULTS: REPRODUCIBLE STEERING AND PROMPT-CONDITIONED COMPETITION

### 4.1 A NON-GENERIC WRITE EFFECT REPRODUCES

The first finding starts with a successful intervention: Qwen Effusion reproduces beyond random and sham controls before clinical alternatives change its attribution.

Table 1 evaluates controlled decodability at the exact final visual block: `CLIP.encoder.layers.22` for LLaVA and `model.visual.blocks.31` for Qwen. Its intervals are 95% patient-cluster bootstrap intervals over 2,000 draws. The control column reports mean AUROC over 20 recurring-type random-label maps and their 5th–95th percentile spread.

Table 1: Controlled decodability of clinical concepts at the final visual block.

| Model         | Concept  | Locus    | AUROC<br>[95% CI]          | Control mean<br>(5–95%)   | Selectivity<br>[95% CI]    |
|---------------|----------|----------|----------------------------|---------------------------|----------------------------|
| LLaVA-1.5-7B  | Effusion | CLIP L22 | 0.7788<br>[0.7538, 0.8031] | 0.6649<br>(0.5750–0.7450) | 0.1139<br>[0.0875, 0.1404] |
| LLaVA-1.5-7B  | Edema    | CLIP L22 | 0.8009<br>[0.7648, 0.8329] | 0.6649<br>(0.5750–0.7450) | 0.1360<br>[0.0994, 0.1693] |
| Qwen2.5-VL-7B | Effusion | Qwen B31 | 0.7742<br>[0.7488, 0.7993] | 0.6576<br>(0.5738–0.7476) | 0.1166<br>[0.0897, 0.1428] |

Qwen Effusion first satisfies the reader test. Its probe reaches AUROC 0.7742 and controlled selectivity 0.1166 (95% CI [0.0897, 0.1428]). Original-cohort intervention outcomes maximize the concept-consistent effect over six fixed nonzero doses. Qwen makes a strong write-direction case in the original 200-image cohort: the maximum concept-consistent change is 0.2343 at  $\alpha = +0.25$ , above the random-direction 95th percentile of 0.0963 and the maximum absolute sham effect of 0.0852.

The response at the locked dose  $\alpha = +0.25$  reproduces on the patient-disjoint 400-patient cohort (Figure 2). Effusion changes mean  $P(\text{yes})$  by 0.1945, again above the random-direction 95th percentile of 0.0878 and absolute sham effect of 0.0157. None of these patients appears in the original intervention cohort, and the dose was not reselected, supporting a large and reproducible non-generic effect within ChestX-ray14.

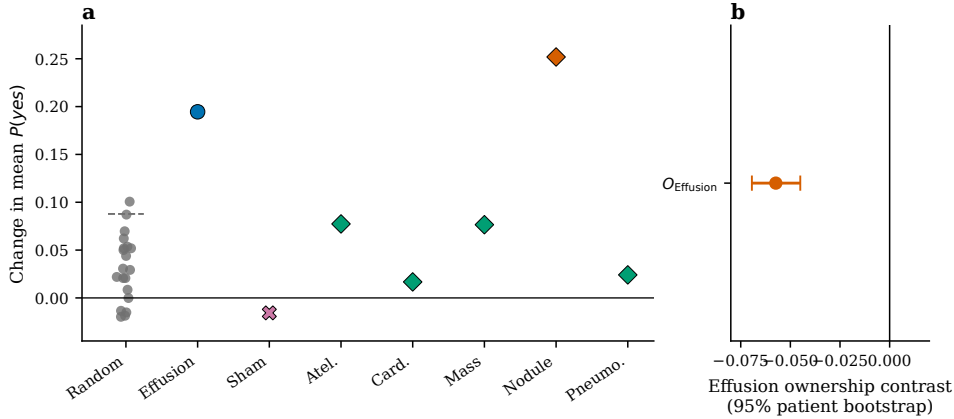


Figure 2: **Qwen Effusion reproduces as a non-generic write direction, but semantic competition changes its attribution.** (a) Changes in mean  $P(\text{yes})$  at the locked dose  $\alpha = +0.25$  on 400 patients absent from the original cohort. The random group contains 20 directions; its dashed segment marks  $p_{95}$ . Diamonds show the five fixed alternative clinical normals. (b) The Effusion ownership contrast  $O_{\text{Effusion}}$  with its two-sided 95% patient-bootstrap interval; the registered one-sided 95% lower bound is  $-0.0678$ . Nodule realizes the point-estimate maximum.

#### 4.2 SEMANTIC COMPETITION CHANGES ATTRIBUTION, NOT EFFICACY

The fixed clinical family provides a harder identification test than non-semantic controls. Nodule changes the Effusion answer by 0.2519 and realizes the point-estimate maximum among the five alternatives. The Effusion ownership contrast  $O_{\text{Effusion}}$  is  $-0.0573$ . The registered fixed-family endpoint recomputes the maximum inside each of 5,000 patient-bootstrap draws, giving a two-sided 95% CI of  $[-0.0694, -0.0450]$  and a one-sided 95% lower bound of  $-0.0678$ . The entire interval is below zero. Descriptively, all five alternative clinical normals move the Effusion endpoint in the same positive direction; Effusion exceeds four by 0.1172–0.1778, while Nodule alone has a larger point estimate. This shared-sign pattern motivates an answer-sensitive-alias hypothesis but cannot distinguish it from question-specific ownership.

Qwen Effusion therefore remains a reproducible direction that changes the answer beyond random and sham perturbations, while its specific semantic attribution fails. The familywise endpoint establishes that at least one fixed clinical alternative is stronger; Nodule is the point-estimate maximizer, not a separately familywise-identified semantic owner.

#### 4.3 WORDING CHANGES CLINICAL COMPETITION WITHIN THE ANSWER INTERFACE

The second finding concerns the conditions of clinical competition. The Qwen ownership pattern persists in logit coordinates (Appendix G), so we test whether changing the answer interface changes a direction’s relative advantage.

In the registered crossover, Atelectasis, Pneumothorax, Mass, and Nodule pass independent capability calibration. On these four questions, mapping-even minus mapping-odd response energy is  $-2.1786$  (95% CI  $[-2.4061, -1.9810]$ ): effects are larger in the component that reverses raw A-minus-B sign when the finding-to-letter mapping is exchanged. Yet all 20 random directions and the sham are also mapping-odd dominant (Appendix H). Aggregate mapping response therefore does not distinguish the clinical normals from generic perturbations.

The crossover identifies Mass as a prompt-conditioned candidate: its clinical margin changes from  $-0.7181$  under the original yes/no prompt to  $0.2825$  and  $0.2900$  under A-present and B-present, respectively (Table A15). A-present assigns A to presence and B to absence; B-present reverses those assignments. Both coded effects exceed the discovery study’s 20 random directions and sham. Because wording and instructions also change, we test them separately on independent patients.

Table 2 reports the registered confirmation on 200 new patients. Clinical margins subtract the largest of five alternative clinical effects in finding-present logit units. Minimum LCB is the smallest simultaneous lower bound among a cell’s five contrasts, from the joint 20-comparison family. All seven prompt cells pass separate capability calibration.

Table 2: Independent Mass competition under coded prompts.

| Condition              | Clinical margin | Minimum LCB | Mass effect | Random max |
|------------------------|-----------------|-------------|-------------|------------|
| <i>show</i> /A-present | 0.2525          | 0.1675      | 1.7019      | 2.2100     |
| <i>show</i> /B-present | 0.2338          | 0.1462      | 1.8188      | 2.2169     |
| <i>is</i> /A-present   | $-0.1087$       | $-0.1751$   | 1.9225      | 2.4750     |
| <i>is</i> /B-present   | $-0.2006$       | $-0.2686$   | 2.0019      | 2.4944     |

The clinical advantage reproduces under both *show* coded prompts, but the Mass effects remain below the maxima of 119 random directions, with reference ranks of 3/120. Under *is*, both clinical margins are negative. Neither the cross-wording nor original-wording specificity conjunction is met. Explicit *show* yes/no also has a positive clinical margin,  $0.2369$ ; its paired differences from A-present and B-present span zero. Within each response instruction, changing *show* to *is* reduces the clinical margin (Table A18). These paired wording contrasts are descriptive. They support wording-conditioned clinical competition, rather than an advantage attributable to answer letters alone. Mass steering increases Brier error in every cell, while all AUROC-change intervals span zero (Tables A20 and A19).

#### 4.4 LIMITS OF CAUSAL AND DIAGNOSTIC INTERPRETATION

**Ownership, shared responses, and input closure.** The broader Qwen tests examine how far the attribution result extends at the same consumed block. Table 3 summarizes the registered comparisons; LCB denotes a one-sided simultaneous or familywise 95% lower confidence bound.

All six diagonal ownership margins are negative in the six-direction by six-question matrix. Effusion’s mean off-diagonal effect,  $0.2909$ , makes it the strongest shared-alias candidate, but the random alias maximum is larger at  $0.3402$ . The matrix therefore identifies neither a replacement clinical owner nor a shared alias under the registered comparisons (Appendix C). On 50 fresh within-patient Consolidation pairs, writing the measured displacement gives mean closure gain  $0.0022$ , below the random maximum of  $0.0025$ . Negative displacement and clinical-comparison lower bounds leave the input-closure conjunction unmet (Appendix D).

Table 3: Prospective tests of clinical ownership, shared aliasing, and input closure in Qwen.

| Gate          | Cohort       | Target statistic       | Registered comparator | Uncertainty boundary                   | Outcome             |
|---------------|--------------|------------------------|-----------------------|----------------------------------------|---------------------|
| Ownership     | 400 patients | Alias candidate 0.2909 | Random max 0.3402     | Alias LCB 0.2808<br>Disp. LCB -0.0364; | 0/6 owned; no alias |
| Input closure | 50 pairs     | Gain 0.0022            | Random max 0.0025     | clin. LCB -0.0015                      | No closure          |

**LLaVA reader and answer measurements.** The original LLaVA Effusion and Edema readers have AUROCs 0.7788 and 0.8009 with positive selectivity intervals (Table 1), but neither clears the write-control conjunction. Effusion’s maximum effect, 0.1470, is below absolute sham 0.1546; Edema’s 0.0659 lies inside its random interval (Figure A7; Table A10). Paired decoding contrasts leave reader ordering unresolved (Table A11).

A 700-patient pilot admits Effusion and Cardiomegaly readers but no clean answer margin, leaving its reserved write cohort unevaluated (Appendix K). Finite text calibration selects yes/no, followed by independent validation on 700 index patients and 700 disjoint donors. Image-linked opportunity and wording dependence remain unresolved, and frozen-reader selectivity does not replicate:  $-0.0111$  (95% CI  $[-0.1541, 0.1185]$ ). The descriptive clinical AUROC change, 0.8196 to 0.6664, accompanies similar control means, 0.6758 and 0.6775. The sampling frames differ, and no between-cohort decline is established (Appendix L).

**Natural paired response.** A separate cohort of 100 fresh Consolidation-positive/negative patient pairs matches view, sex, integer age, and the other 13 disease labels. Both models evaluate the same images under the original yes/no prompt, with all patients and the same 5,000 bootstrap draws retained. Table 4 reports the mean positive-minus-negative answer-logit gap and concordance, which gives half credit to zero gaps.

Table 4: Paired Consolidation response on a shared cohort.

| Model         | Mean paired gap | One-sided 95% lower bound | Concordance |
|---------------|-----------------|---------------------------|-------------|
| Qwen2.5-VL-7B | -0.1175         | -0.3987                   | 0.545       |
| LLaVA-1.5-7B  | -0.0712         | -0.1338                   | 0.455       |

Neither model meets the positive-lower-bound opportunity criterion. Their direct patient-paired concordance difference, LLaVA minus Qwen, is  $-0.090$  (descriptive 95% CI  $[-0.210, 0.025]$ ). Positive paired opportunity and the model difference remain unresolved; raw gaps retain their within-model coordinates (Appendix J). These measurements constrain the interpretation of input closure without identifying its mechanism.

## 5 DISCUSSION AND LIMITATIONS

The Qwen Effusion result changes the attribution question after steering succeeds. Its locked-dose response reproduces beyond random and sham controls, yet a fixed clinical alternative produces a larger change on the same patients and endpoint. Non-semantic controls establish an effective write direction; matched clinical competition tests whether its training label grants it relative privilege. The negative ownership interval answers the latter comparison without identifying which clinical concept the model uses.

Mass shows why this comparison also needs a specified prompt. Its advantage over all five clinical alternatives reproduces under both *show* A/B instructions, while both *is* margins are negative. Explicit *show* yes/no has a similar observed margin, and the paired wording contrasts support a contribution from wording rather than answer letters alone. Clinical advantage and specificity remain distinct: larger effects among 119 random directions defeat the full specificity criteria.

These findings concern control of an answer, not improved diagnosis. Across Mass confirmation prompts, all AUROC-change intervals span zero and Brier error increases. Under *is*, clinical margins



improve from the original anchor to explicit yes/no to A/B instructions, but that change in relative competition does not translate into demonstrated diagnostic benefit.

**Limitations.** The independent findings condition on ChestX-ray14, one Qwen visual locus, a fixed direction bundle, and the evaluated doses and prompts. Patient-bootstrap intervals quantify uncertainty over patients, not retraining of directions. Mass confirmation has 12 positives among 200 patients. Paired prompt contrasts are descriptive, and intervals spanning zero do not establish equivalence. The discovery crossover reused 200 patients and 20 random controls; its energy endpoint covers four calibration-eligible questions. Report-derived labels, acquisition correlations, and mean pooling limit clinical interpretation.

The matrix, closure, and LLaVA analyses delimit the causal interpretation (Section 4.4). They establish neither a shared answer mechanism nor reproducible reader–answer separation. Pilot intervals are unadjusted for selection or multiplicity, and the independent reader comparison uses a different sampling frame (Appendices K and L). Metadata-matched longitudinal pairs retain unrecorded differences; their opportunity and direct model contrast remain unresolved. The cross-model comparison covers native configurations, including training and preprocessing, with half-credit ties under evaluation configured for BF16 (Appendix J).

## 6 CONCLUSION

Successful steering leaves label-specific attribution open to semantic competition. Within the fixed NIH ChestX-ray14 direction bundle and Qwen visual locus, Effusion replicates beyond random and sham controls yet loses to a clinical competitor. Mass gains clinical advantage under both *show* A/B mappings and loses it under *is*, while larger random effects defeat specificity. These findings separate intervention efficacy from relative clinical advantage and expose its wording dependence. The measured advantage establishes neither a mechanism of concept use nor diagnostic benefit.

### AI USE STATEMENT

Generative AI tools assisted with hypothesis refinement, experimental-design feedback, analysis-code implementation and review, interpretation of experimental outputs, figure creation, literature discovery, and drafting and editing the manuscript. All reported experiments were executed through the documented protocols; numerical claims were checked against saved artifacts, the analysis code was tested, and the authors reviewed all AI-assisted content. The authors take responsibility for the final content of this work, including text, claims, code, and artifacts produced with AI assistance.

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## A PROMPTS, SCORES, AND NOTATION

The appendix develops the evidence behind the reader–writer comparison. This section specifies the original endpoint and notation; Appendix B identifies the cohorts and computation behind each experiment. Appendices C and D examine semantic competition and paired input closure in detail. Appendix E checks that the interventions enter the consumed forward path and that their results replay. Appendix F compares the original intervention outcomes and reader selectivity. Appendices G and H then examine score-coordinate diagnostics and answer-encoding sensitivity; Appendix I tests the Mass candidate on independent patients across question wordings and response instructions. Appendix J compares paired Consolidation response on a shared cohort across two model configurations. Appendix K reports LLaVA’s pilot reader and answer screen. Appendix L follows it with text calibration and independent image/reader validation.

### A.1 A COMMON QUESTION AND ANSWER ENDPOINT

At the first answer position, let  $\ell_{\text{yes}}$  and  $\ell_{\text{no}}$  denote the maximum log probabilities among the valid single-token case and leading-space variants of each answer. The registered score is  $P(\text{yes}) = \sigma(\ell_{\text{yes}} - \ell_{\text{no}})$ , where  $\sigma$  is the logistic sigmoid. Figure A1 shows how the six ownership questions share their wording and scoring pipeline: only the finding phrase changes. The original Edema cell uses “pulmonary edema,” and the closure experiment uses “consolidation” in the same template. Taking the best valid token variant for each answer makes the endpoint insensitive to which listed capitalization or leading-space form receives the highest score. This endpoint measures the relative support for yes versus no at the first answer position; it does not measure the accuracy of a generated clinical report.

#### Fixed clinical question

Is there <finding> in this chest radiograph?  
Answer yes or no.

#### Six ownership findings

Only the finding phrase changes

|              |                    |              |               |
|--------------|--------------------|--------------|---------------|
| Effusion     | a pleural effusion | Cardiomegaly | cardiomegaly  |
| Atelectasis  | atelectasis        | Mass         | a lung mass   |
| Pneumothorax | a pneumothorax     | Nodule       | a lung nodule |

#### First-answer scoring

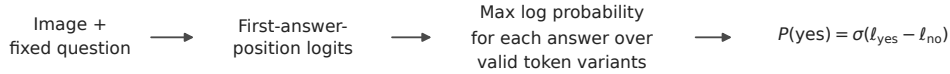


Figure A1: **The registered prompts share one template and one answer score.** The six findings instantiate an otherwise fixed yes/no question, and every model uses the same first-answer-position scoring sequence.

The shared interface makes the question–direction grid interpretable: changing a clinical direction does not change the question being scored. Differences across columns still reflect different clinical questions, so the ownership test compares directions within each column before drawing a conclusion about the named concept.

### A.2 READING THE INTERVENTION NOTATION

Table A1 collects the notation introduced in the article. The distinction between question  $q$  and direction  $d$  is central: a large  $W_{q,d}$  shows that a direction moves an answer, while  $O_q$  asks whether the direction bearing that answer’s clinical name moves it more than its competitors. A positive ownership margin alone is not the full confirmation rule; the uncertainty and non-semantic control

comparisons must also pass. Keeping these roles separate allows the following analyses to distinguish readability, intervention strength, and semantic attribution.

Table A1: Notation used in the manuscript.

| Symbol               | Definition                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------|
| $q, d, c$            | Clinical question, steering direction, and probe concept, respectively.                           |
| $\mathbf{h}_{it}$    | Consumed visual representation for image $i$ and visual token $t$ .                               |
| $R, s$               | Fixed Gaussian projection and train-fitted featurewise scale.                                     |
| $\hat{\mathbf{w}}_c$ | Unit probe-normal direction for concept $c$ after lifting to the consumed representation.         |
| $\alpha$             | Signed intervention dose; $ \alpha $ is the perturbation norm relative to each token norm.        |
| $W_{q,d}$            | Paired mean change in question $q$ 's $P(\text{yes})$ under direction $d$ .                       |
| $O_q$                | Ownership margin $W_{q,q} - \max_{d \neq q} W_{q,d}$ ; positive values favor the named direction. |
| $S$                  | Real-label AUROC minus the mean AUROC of the 20 nuisance-type control tasks.                      |
| $\gamma_i$           | Positive-minus-negative paired representation displacement along the Consolidation normal.        |

The notation separates three levels of evidence:  $S$  describes label availability,  $W_{q,d}$  describes an intervention effect, and  $O_q$  compares the named direction with clinical alternatives. The remaining appendix follows this progression while keeping the evaluation cohorts explicit.

## B REGISTERED RUNS AND PROVENANCE

### B.1 EVALUATION UNITS AND COMPUTATION

Table A2 connects each experimental question to its evaluation units and computational budget. R1–R3 evaluate probes on the held-out patient split and interventions on 200 images per cell; these are different evaluation sets. R4 tests the locked Effusion dose on 400 independent patients, R5 crosses questions and clinical directions on a further 400-patient cohort, and R6 uses paired studies from 50 fresh patients. The ownership experiment accounts for 9,211 of the 20,939 recorded GPU seconds because it evaluates the full question–direction surface with 119 random directions. These are recorded run times, including registered reuse of saved computations, rather than a hardware-normalized cost comparison. Each run used one A100.

Table A2: Experimental runs and computational budgets.

| Run | Model | Endpoint              | Evaluation units        | Random | Bootstraps | GPU s |
|-----|-------|-----------------------|-------------------------|--------|------------|-------|
|     |       |                       | 1,659 probe patients    |        |            |       |
| R1  | LLaVA | Effusion cell         | 200 intervention images | 20     | 2,000      | 7,022 |
|     |       |                       | 1,659 probe patients    |        |            |       |
| R2  | LLaVA | Edema cell            | 200 intervention images | 20     | 2,000      | 1,823 |
|     |       |                       | 1,659 probe patients    |        |            |       |
| R3  | Qwen  | Effusion cell         | 200 intervention images | 20     | 2,000      | 2,077 |
| R4  | Qwen  | Effusion specificity  | 400 patients            | 20     | 5,000      | 361   |
| R5  | Qwen  | Causal ownership      | 400 patients            | 119    | 5,000      | 9,211 |
| R6  | Qwen  | Consolidation closure | 50 patient pairs        | 119    | 5,000      | 445   |

The progression from single cells to the full clinical grid broadens the competitor family without changing the basic answer endpoint. The final paired experiment then asks a different question: whether the direction tracks and closes a within-patient input difference. Its 50 pairs must therefore be interpreted separately from the unpaired 400-patient tests.

### B.2 COHORT IDENTITY AND FIXED SETTINGS

Table A3 links the run labels to immutable executions and gives the row or pair selection hashes for the prospective cohorts. The specificity cohort excludes patients in the original intervention set, and ownership uses a third label-blind cohort. Closure excludes the earlier Qwen evaluation rows and selects one positive and one negative study for each eligible patient. These hashes identify the exact evaluation units used in those comparisons, so a replay can be tied to the same cohort as the reported result.

Table A3: Immutable run identifiers and evaluation-cohort selection hashes.

| Run | Run identifier and cohort selection hash                                                                                              |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|
| R1  | 20260902T191411Z-ffd523c464c8-f99e2f39                                                                                                |
| R2  | 20260902T221534Z-110b84618d1b-edema                                                                                                   |
| R3  | 20260903T000321Z-caaae3ef346d-qwen-full<br>20260903T103508Z-456c81bad460-7a3edc4b; row SHA-256                                        |
| R4  | 76be1da9659973d44a0898fa69c579afc4649f329ee227b19d1216efaa1c8bda<br>20260904T125710Z-a3bd883540eb-causal-ownership; row SHA-256       |
| R5  | e657a9bd1707f13695b02a341047af6f53981295b840376eb0ea5bbefbe2d312<br>20260904T162317Z-8a55a4c2f6b6-consolidation-closure; pair SHA-256 |
| R6  | 20dcc8cc3aaa8c65a6f7f759a0ff6bf3b0cf4f1b87f4b65cdc6a15a46c6d85e2                                                                      |

These identifiers preserve the link between an estimate and the patients who produced it. They also distinguish the two 400-patient experiments: specificity evaluates the Effusion endpoint, whereas ownership evaluates all six questions on its own cohort. Similar sample sizes do not make their numerical effects interchangeable.

Table A4 records which choices remain fixed when moving from decoding to intervention. The model revisions, projection, and readout settings determine the clinical normals; the dose grids and control families determine the perturbation comparisons. In particular, the original target curve has nine doses, whereas random and sham comparisons use six nonzero doses. The prospective ownership test fixes the dose at  $\alpha = +0.25$ , while closure derives a dose from each measured input pair. Those designs address different questions and should be read with their corresponding control rules.

Table A4: Shared registered settings.

| Component          | Registered value                                                                                                |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| LLaVA model        | llava-hf/llava-1.5-7b-hf                                                                                        |
| LLaVA revision     | b234b804b114d9e37bb655e11cbbb5f5e971b7a9                                                                        |
| Qwen model         | Qwen/Qwen2.5-VL-7B-Instruct; processor resolution 336 <sup>2</sup>                                              |
| Qwen revision      | cc594898137f460bfe9f0759e9844b3ce807cfb5<br>512-dimensional Gaussian projection; logistic regression, $C = 1$ ; |
| Readout            | projection and fit seed 0                                                                                       |
| Control labels     | 20 recurring-type maps, seeds 0–19                                                                              |
| Original dose grid | -1, -0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5, 1                                                                     |
| Controlled doses   | -1, -0.5, -0.25, 0.25, 0.5, 1                                                                                   |
| Generic controls   | Isotropic random directions and a coordinate-permutation sham                                                   |
| Clinical family    | Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, Nodule                                                 |
| Ownership          | $\alpha = +0.25$ ; $6 \times 6$ grid; 119 random directions; seed 20260904                                      |
| Input closure      | 50 pairs; displacement-matched dose; 119 random directions; seed 20260905                                       |

Fixing the readout construction makes the clinical directions comparable as candidates, but it does not guarantee their specificity as writers. The larger random family in ownership and closure also imposes a stronger maximum-control comparison than the original p95 rule. The next analyses interpret each result against the rule registered for that experiment.

### B.3 FROM REGISTERED DESIGN TO STATISTICAL DECISIONS

Figure A2 connects the experimental design to the patient-level estimates and their statistical interpretation. Model, visual locus, dose rule, and cohort selection are specified before each test. Ownership and aliasing use 400 patients and 304,800 evaluations; closure uses 50 patient pairs and 6,450 evaluations. Evaluation counts include the perturbation conditions applied to each study, while the bootstrap resamples patients or within-patient pairs. Ownership and alias decisions combine patient-bootstrap bounds with the registered clinical, random, and sham comparisons. Closure requires the displacement and gain comparisons to satisfy the registered conjunction.

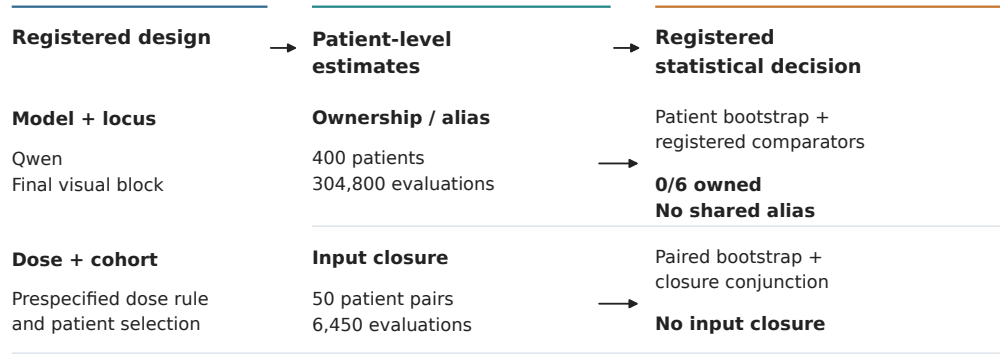


Figure A2: **Registered designs connect patient-level estimates to statistical decisions.** The two experiments evaluate different patient cohorts under prespecified dose rules. Their bootstrap and comparator requirements yield zero named owners, no detected shared alias, and no confirmed input closure.

The decision step distinguishes an observed effect from a supported causal interpretation. Neither the six named ownership tests nor the shared-alias test meets its confirmation rule, and the paired experiment does not confirm input closure. These conclusions depend on the patient-level uncertainty and comparator results, rather than on the number of evaluations. The full ownership surface below shows which clinical competitors drive the first of these outcomes.

## C COMPLETE CAUSAL-OWNERSHIP SURFACE

### C.1 QUESTION-SPECIFIC EFFECTS AND CLINICAL COMPETITORS

Here  $q$  indexes a clinical question,  $d$  indexes a steering direction, and  $W_{q,d}$  retains its definition from Section 3. Rows in Figure A3 are directions and columns are questions, so the named comparison lies on the diagonal. The column-wise ownership margin  $O_q$  is positive only when that diagonal exceeds every clinical competitor. The off-diagonal maxima reveal why a strong write effect does not identify its clinical owner. For example, the Mass answer changes by 0.511 under the Effusion direction versus 0.359 under its own direction, and the Effusion answer changes by 0.245 under Nodule versus 0.181 under Effusion. All six columns have an off-diagonal winner, so the mismatch extends beyond the original Effusion comparison.

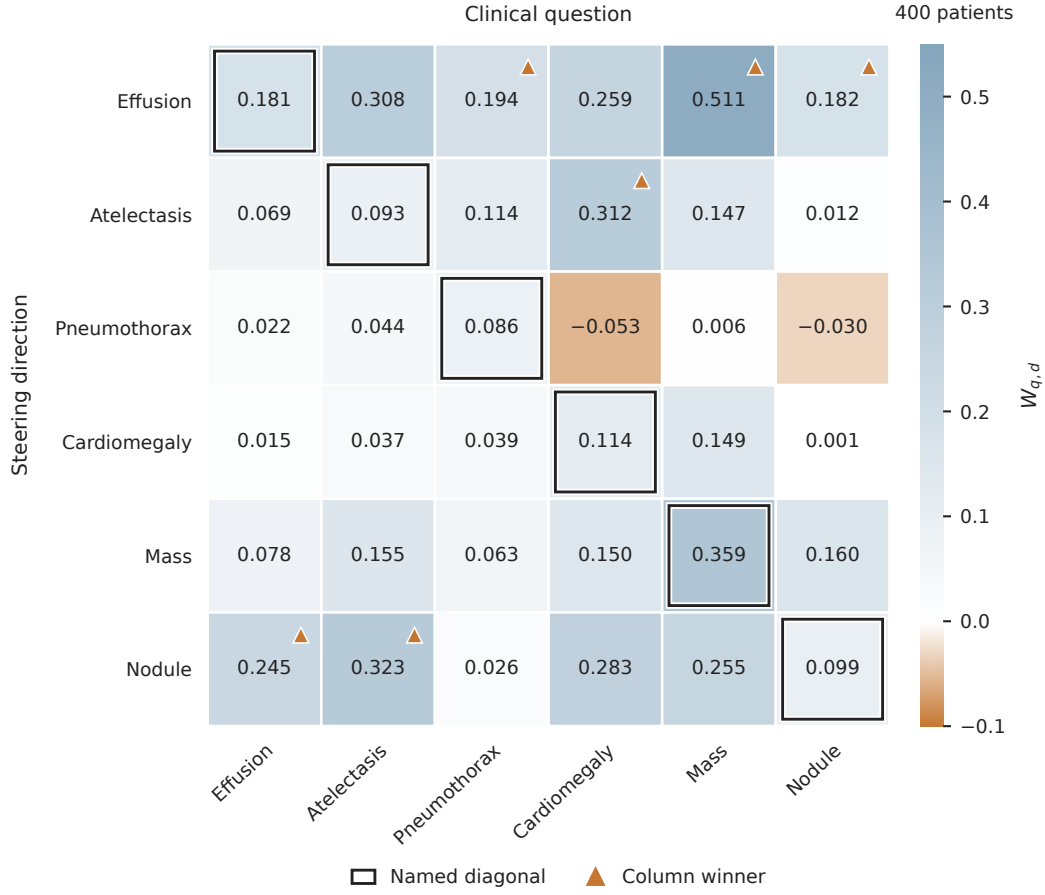


Figure A3: **Every clinical question is moved most by an off-diagonal direction.** Outlined cells mark the named diagonals, and orange triangles mark each column’s strongest clinical direction. The continuous color scale encodes signed effects, with white at zero; printed values give the corresponding estimates to three decimal places.

The surface shows substantial answer movement alongside a systematic failure of diagonal dominance. Several questions share the same strongest direction, which motivates both the per-question ownership test below and the subsequent test for a shared alias. A visually strong row alone cannot settle either claim.

Table A5 quantifies these comparisons with patient-bootstrap uncertainty. “Best other” is the strongest alternative clinical effect for the question in that row; “Sim. LCB” reports the simultaneous lower confidence bound for  $O_q$ , using the max- $T$  procedure over the clinical contrasts. The descriptive intervals also apply to  $O_q$ . The reported margins range from  $-0.230$  to  $-0.064$ , and every descriptive 95% interval lies below zero. The named diagonals also fall below their corresponding random maxima. Thus neither clinical competition nor the random-control comparison supports a named owner. The displayed effects and margins are rounded independently from the full-precision estimates.

Table A5: Per-question ownership statistics.

| Question $q$ | $W_{q,q}$ | Best other | Winner $d$  | $O_q$ | 95% CI         | Sim. LCB | Random max |
|--------------|-----------|------------|-------------|-------|----------------|----------|------------|
| Atelectasis  | .093      | .323       | Nodule      | -.230 | [-.238, -.221] | -.242    | .367       |
| Cardiomegaly | .114      | .312       | Atelectasis | -.197 | [-.208, -.186] | -.213    | .521       |
| Effusion     | .181      | .245       | Nodule      | -.064 | [-.075, -.053] | -.080    | .264       |
| Mass         | .359      | .511       | Effusion    | -.153 | [-.164, -.141] | -.169    | .611       |
| Nodule       | .099      | .182       | Effusion    | -.083 | [-.096, -.071] | -.101    | .214       |
| Pneumothorax | .086      | .194       | Effusion    | -.109 | [-.116, -.102] | -.119    | .475       |

All six named ownership tests fail. The point-estimate winners identify where the largest observed clinical effects lie, but the table does not establish those alternative directions as semantic owners. A direction that moves several answers could instead reflect a shared response axis, which requires its own control comparison.

## C.2 TESTING A SHARED ANSWER-SENSITIVE DIRECTION

Figure A4 asks whether a direction that lacks a specific clinical owner nevertheless acts as a shared answer-sensitive alias. Panel (a) compares signed mean off-diagonal effects across directions; panel (b) places the strongest candidate against the global controls. Effusion has mean effect 0.2909 and positive simultaneous lower bounds for effect and relative dominance, 0.2808 and 0.1332. Its candidate effect still falls below the global random alias maximum of 0.3402. This control first averages each random direction’s effect over each five-question off-diagonal set, then takes the maximum over all 119 directions and six sets. It is therefore distinct from the individual-question maxima in Table A5. The sham control is the maximum absolute mean effect across individual questions. Because alias confirmation requires all of these comparisons to pass, the broad response pattern does not confirm the shared-alias hypothesis.

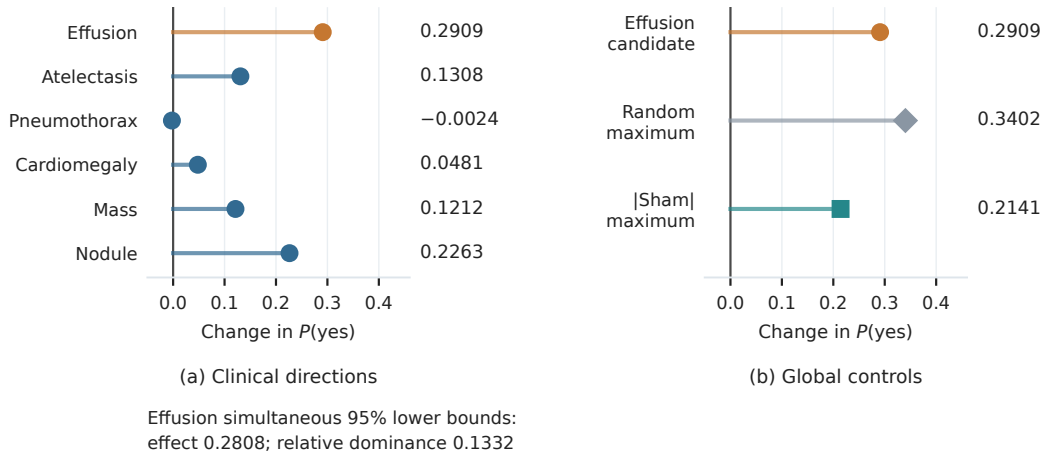


Figure A4: **Effusion is the strongest shared-alias candidate but remains below the registered random-control maximum.** Both panels use a common  $P(\text{yes})$ -change axis and zero baseline. Markers indicate point estimates, with horizontal stems showing their signed distance from zero. Simultaneous max- $T$  lower bounds for the candidate are reported separately.

Effusion’s broad effect therefore remains a candidate pattern rather than a confirmed alias. The random family contains a stronger effect under the registered comparison, so the matrix supports neither named ownership nor the proposed shared alias. The following section relates this result to the earlier specificity test and the paired-input fallback.

## D DIRECTION SPECIFICITY AND INPUT CLOSURE

### D.1 REPLICATION UNDER A FIXED CLINICAL COMPETITOR FAMILY

Table A6 separates a reproducible write effect from its semantic attribution. On the independent 400-patient cohort, Effusion’s effect of 0.1945 exceeds random p95 and the absolute sham effect, but Nodule reaches 0.2519 at the same locked dose. The resulting ownership contrast is  $-0.0573$ , with its entire two-sided interval below zero. The bottom group records the original 200-image cohort, where Nodule also had the larger descriptive effect; those values motivated the prospective test and are not pooled with its estimates. This comparison supports replication of a non-generic intervention effect while rejecting Effusion-specific attribution under the registered competitor family.



Table A6: Locked-dose Qwen Effusion specificity.

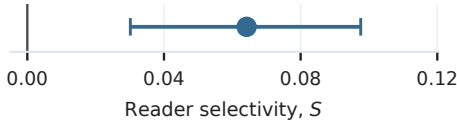
| Direction or control                                   | Effect | Direction or control     | Effect           |
|--------------------------------------------------------|--------|--------------------------|------------------|
| Effusion target                                        | .1945  | Random-direction p95     | .0878            |
| Nodule                                                 | .2519  | [Sham]                   | .0157            |
| Atelectasis                                            | .0773  | Cardiomegaly             | .0167            |
| Mass                                                   | .0765  | Pneumothorax             | .0241            |
| $O_{\text{Effusion}}$                                  | -.0573 | 95% patient-bootstrap CI | [-.0694, -.0450] |
| <i>Original 200-image cohort (descriptive context)</i> |        |                          |                  |
| Nodule at the same dose                                | 0.2863 | Effusion target          | 0.2343           |

The independent cohort preserves the intervention’s strength while exposing its attribution problem: the named Effusion direction is not privileged for the Effusion answer. This distinction motivates testing whether an eligible clinical direction aligns with an actual paired input change, rather than inferring that alignment from a large write effect.

## D.2 FROM PAIRED INPUT DISPLACEMENT TO CLOSURE

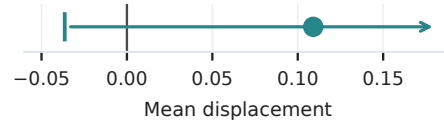
For the paired closure gate,  $\gamma_i$  is the positive-minus-negative representation displacement for patient  $i$ , and  $\alpha_i$  is the displacement-matched dose applied to that patient’s negative study. The closure gain measures the reduction in the absolute paired  $P(\text{yes})$  gap. These quantities are separated into distinct panels in Figure A5 because they have different scales. Panel (a), reader eligibility, gives Consolidation selectivity 0.0642 (95% CI [0.0302, 0.0976]) before pair selection. Panel (b) shows mean paired displacement 0.1091 with one-sided lower bound  $-0.0364$ . The closure-gains panel (c) shows that the concept’s gain of 0.00220 is below the random maximum of 0.00246; the clinical-margin panel (d) adds a negative lower bound for the clinical contrast. The random-rank  $p$ -value of 0.025 is descriptive and does not replace the registered requirement to exceed every random direction. The joint closure criterion therefore remains unmet despite the eligible reader and positive point estimates.

AUROC 0.7218; selectivity 0.0642  
95% CI [0.0302, 0.0976]

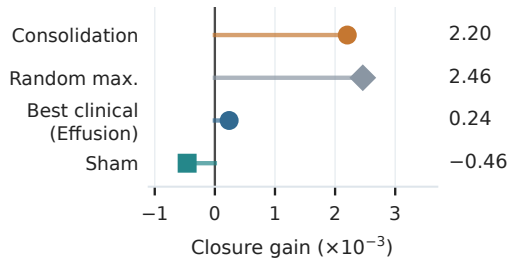


(a) Reader eligibility

50 patient pairs; estimate 0.1091  
95% lower bound  $-0.0364$

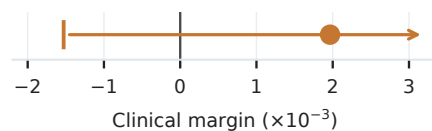


(b) Paired displacement



(c) Closure gains

Estimate 1.97; 95% lower bound  $-1.53$   
Exact random-rank  $p = 0.025$  (descriptive)



(d) Clinical margin

Figure A5: **The Consolidation direction is readable, but its measured write does not reliably close the paired input gap.** Panel (a) shows the reader estimate and two-sided 95% interval. Panels (b) and (d) show point estimates and one-sided 95% lower bounds; arrows extend toward larger values rather than indicating a finite upper confidence limit. Panel (c) compares closure gains on a common axis with a zero baseline. Panels (c) and (d) display values in units of  $10^{-3}$ ; the descriptive random-rank result does not replace the joint confirmation rule.

The failure occurs at the link between readable information and a reliable paired causal effect. A positive reader-selectivity interval establishes availability, but neither the displacement uncertainty nor the registered gain comparisons confirms input closure. The dose diagnostics below help interpret this outcome under the measured-displacement design.

Table A7 checks how the displacement-matched doses relate to the registered cap. The minimum, median, and maximum describe the signed dose distribution, while the clipped fraction tests whether the allowed  $[-0.25, 0.25]$  interval truncates the measured displacements. The final row identifies the strongest alternative clinical gain and the sham response, completing the control comparison in Figure A5.

Table A7: Input-closure dose and control diagnostics.

| Quantity               | Value                   | Quantity            | Value                  |
|------------------------|-------------------------|---------------------|------------------------|
| Random directions      | 119                     | Bootstrap resamples | 5,000                  |
| Minimum $\alpha_i$     | $-3.653 \times 10^{-4}$ | Median $\alpha_i$   | $3.558 \times 10^{-5}$ |
| Maximum $\alpha_i$     | $4.067 \times 10^{-4}$  | Fraction at cap     | 0                      |
| Maximum unrelated gain | .000238 (Effusion)      | Sham gain           | -.000461               |

All doses lie between approximately  $-3.65 \times 10^{-4}$  and  $4.07 \times 10^{-4}$ , with no clipping, so the cap does not explain the failed conjunction. Effusion yields the largest alternative clinical gain, and the sham gain is negative. These small doses characterize the measured displacement; they do not establish what a larger, separately selected write would do. The next section checks that the implemented write reaches the answer computation.

## E FORWARD-PATH AND REPLAY CHECKS

### E.1 DOES THE WRITE REACH THE CONSUMED PATHWAY?

Figure A6 locates the intervention relative to the connector or merger and answer logits. A captured activation must be consumed downstream for its modification to test the model’s behavior. The zero-dose check tests whether installing the hook changes the unperturbed computation; the nonzero-dose checks test whether a write propagates beyond the captured block. The diagram locates both checks along the forward path and shows where their downstream witnesses are measured.

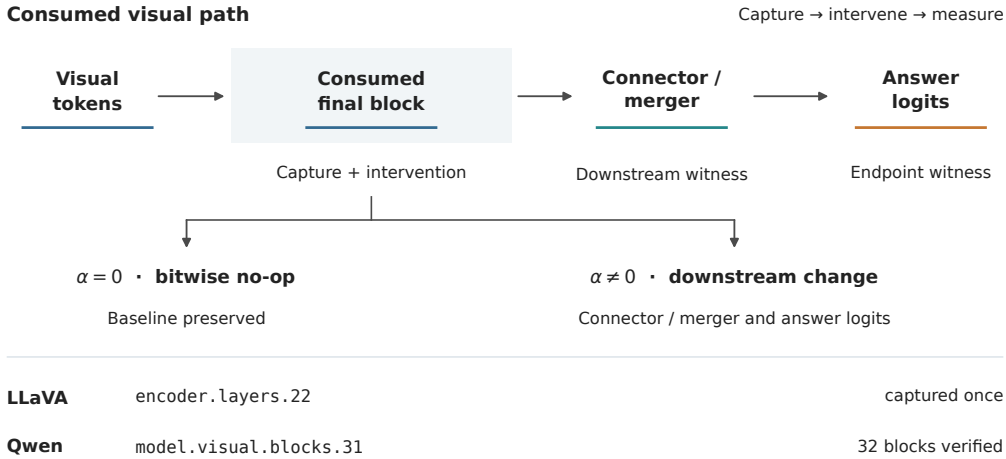


Figure A6: **A valid intervention must enter the consumed forward path and reach the answer logits.** The same hook is tested at zero and nonzero dose before its scientific outcomes are interpreted.

The path checks distinguish a valid manipulation from a change to an unused activation. They make the observed answer effects attributable to the implemented write operation, while leaving the di-

reception’s semantic meaning to the experimental controls. The model-specific witnesses provide the corresponding implementation evidence.

Table A8 gives the corresponding model-specific witnesses. Here the maximum absolute change is taken elementwise between the perturbed and unperturbed downstream tensors or logits. LLaVA’s hook fires once at the registered block and produces nonzero connector and logit changes; Qwen’s final visual block likewise affects both the merger and answer logits at  $\alpha = +0.1$ . The zero-dose column tests baseline preservation, and the final two columns test propagation through the model-specific connector and answer computation.

Table A8: Forward-path intervention checks.

| Model | Captured module                              | Zero dose     | Downstream                          | Logits                    |
|-------|----------------------------------------------|---------------|-------------------------------------|---------------------------|
| LLaVA | encoder.layers.22;                           | bitwise no-op | connector max                       | max $ \Delta  = 0.1875$   |
|       | captured once<br>model.visual.<br>blocks.31; |               | $ \Delta  = 1.1875$                 |                           |
| Qwen  | 32 blocks verified                           | bitwise no-op | merger changed<br>at $\alpha = +.1$ | changed at $\alpha = +.1$ |

For both models, the write reaches the answer computation and the zero-dose operation preserves its baseline. Thus a disconnected hook or an altered zero-dose baseline does not explain the reported reader–writer mismatch. Having established that the manipulation reaches the model output, the remaining computational check is whether its saved outcomes reproduce the reported decisions.

## E.2 CAN THE REPORTED DECISIONS BE REPRODUCED?

Table A9 reports replay at the decision level for the two prospective mechanism gates. Ownership reproduces the mechanism route byte for byte, and closure reproduces all 257 decision arrays from its saved outcomes and bootstrap calculation. This agreement connects the reported conclusions to their saved evidence and complements the forward-path checks. It is a reproducibility check on the reported experiments, rather than an additional independent cohort or replication.

Table A9: Reproducibility of prospective mechanism decisions.

| Gate             | Outcomes | Bootstrap draws | Replay result                  |
|------------------|----------|-----------------|--------------------------------|
| Causal ownership | 304,800  | 5,000           | mechanism route byte-identical |
| Input closure    | 6,450    | 5,000           | 257/257 decision arrays exact  |

Exact replay preserves the negative mechanism conclusions under the saved analysis: no named ownership, no detected shared alias, and no confirmed input closure. It closes the computational evidence chain; the paired reader contrasts below address the remaining possibility that the writer differences simply mirror a resolved reader ranking.

## F ORIGINAL-CELL INTERVENTIONS AND DECODING CONTRASTS

Figure A7 compares the original dose responses. The target effects vary smoothly in some cells, but the random and sham comparisons determine whether the response is unusual for that perturbation scale.

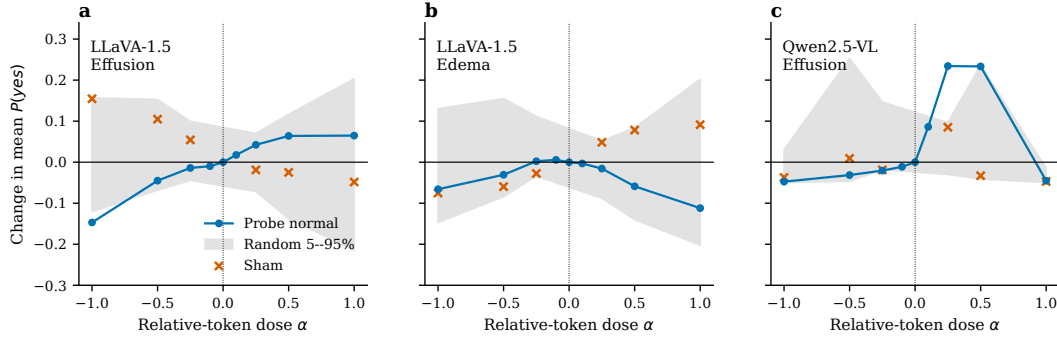


Figure A7: **Comparable readers become different writers at the consumed final visual block.** The blue curve is the raw change in mean  $P(\text{yes})$  under the target probe normal on the same 200 held-out images per cell. Gray bands show the 5th–95th percentile across 20 equal-norm random directions at the six controlled nonzero doses; orange crosses show the coordinate-permutation sham. The concept-consistent rule uses  $\text{sign}(\alpha)\Delta(\alpha)$ , accounting for the negative-dose maxima in the two LLaVA cells.

LLaVA Effusion is centrally monotone but does not exceed the maximum absolute sham response; LLaVA Edema remains inside its random interval. Qwen Effusion passes the original generic-control comparison and reproduces at the locked dose. Table A10 collects these outcomes. Original-cohort effects maximize the concept-consistent response over six fixed nonzero doses; the patient-disjoint supplement fixes  $\alpha = +0.25$ . “Supported interpretation” gives the strongest claim warranted by each cohort.

Table A10: Intervention outcomes and the scientific claims supported by their controls.

| <i>Generic-control outcomes</i> |          |                  |     |          |                 |             |        |
|---------------------------------|----------|------------------|-----|----------|-----------------|-------------|--------|
| Model                           | Concept  | Cohort           | $N$ | $\alpha$ | Target $\Delta$ | Random 95th | sham   |
| LLaVA-1.5-7B                    | Effusion | Original         | 200 | -1.00    | 0.1470          | 0.1202      | 0.1546 |
| LLaVA-1.5-7B                    | Edema    | Original         | 200 | -1.00    | 0.0659          | 0.1470      | 0.0913 |
| Qwen2.5-VL-7B                   | Effusion | Original         | 200 | +0.25    | 0.2343          | 0.0963      | 0.0852 |
| Qwen2.5-VL-7B                   | Effusion | Patient-disjoint | 400 | +0.25    | 0.1945          | 0.0878      | 0.0157 |

| <i>Fixed-family ownership outcome</i> |          |                  |                                |                                  |
|---------------------------------------|----------|------------------|--------------------------------|----------------------------------|
| Model                                 | Concept  | Cohort           | $O_{\text{Effusion}}$ [95% CI] | Supported interpretation         |
| Qwen2.5-VL-7B                         | Effusion | Patient-disjoint | -0.0573 [-0.0694, -0.0450]     | Writer; Effusion ownership fails |

The generic-control and clinical-family rows separate intervention efficacy from its attribution. Qwen’s replicated response exceeds the non-semantic comparators, but the registered familywise clinical margin remains negative.

Table A11 tests whether the different writer outcomes are accompanied by a resolved ordering of reader selectivity. Within LLaVA, Edema exceeds Effusion by 0.0221 in the point estimate; across models, Qwen exceeds LLaVA on Effusion by 0.0026. Both paired patient-bootstrap intervals span zero, so these data do not resolve either ordering. This does not establish equivalence of the readers. It shows that the observed separation in intervention outcomes occurs without a statistically resolved selectivity ranking in these two comparisons, reinforcing the distinction between what a probe can read and what its direction causally controls.

Table A11: Paired controlled-decoding contrasts.

| Contrast                       | Estimate | 95% patient-bootstrap CI |
|--------------------------------|----------|--------------------------|
| LLaVA Edema – LLaVA Effusion   | 0.0221   | [−0.0175, 0.0590]        |
| Qwen Effusion – LLaVA Effusion | 0.0026   | [−0.0155, 0.0215]        |

The reader comparisons therefore do not explain away the writer outcomes through a demonstrated ordering of decoding quality. Taken together, the appendix supports a limited but coherent conclusion: label availability, non-generic answer movement, and concept-specific causal control are distinct empirical properties and require distinct evidence.

## G SCORE COORDINATES AND ANSWER DISCRIMINATION

We retrospectively analyze the stored outputs from the 400-patient Qwen causal-ownership matrix cohort at  $\alpha = +0.25$ , distinct from the earlier 400-patient Effusion specificity supplement. The logit margin is the yes-token logit minus the no-token logit; each intervention effect is its change relative to the same patient’s unperturbed output. In each score coordinate, the ownership margin subtracts the largest mean effect among the five other clinical directions from the matched direction’s mean effect. Table A12 compares these margins with answer discrimination. Here, AUROC ranks patients by the model’s answer logit margin against the corresponding disease labels, rather than by a fitted probe score. The probability-ownership values here belong to the matrix cohort: its Effusion margin is  $-0.0640$ , whereas the earlier specificity supplement reports  $-0.0573$  on different patients.

Table A12: Score-coordinate ownership and answer discrimination.

| Question     | Probability ownership | Logit-margin ownership | Baseline AUROC | Matched-direction AUROC |
|--------------|-----------------------|------------------------|----------------|-------------------------|
| Effusion     | $-0.0640$             | $-0.4853$              | 0.6178         | 0.5921                  |
| Atelectasis  | $-0.2295$             | $-1.0703$              | 0.6235         | 0.5479                  |
| Pneumothorax | $-0.1086$             | $-0.8944$              | 0.6666         | 0.6324                  |
| Cardiomegaly | $-0.1972$             | $-0.8559$              | 0.6217         | 0.5152                  |
| Mass         | $-0.1528$             | $-0.7134$              | 0.6636         | 0.6354                  |
| Nodule       | $-0.0831$             | $-0.4134$              | 0.6805         | 0.6355                  |

All six ownership margins remain negative before the sigmoid mapping to probability. Answer-score AUROC is lower under each matched direction in the point estimates, separating a larger affirmative response from better patient discrimination. These are descriptive comparisons, with the registered ownership decisions retaining their original inferential status.

### G.1 LABEL-CONDITIONED SHIFTS AND MATRIX GEOMETRY

For the Effusion direction on the Effusion question, the mean logit-margin shift is 2.1336 among 29 label-positive patients and 2.2712 among 371 label-negative patients. The positive-minus-negative difference is  $-0.1376$ , with a pointwise 95% patient-bootstrap percentile interval of  $[-0.4443, 0.1659]$  from 2,000 draws. Both groups show an upward shift, while their mean-shift difference remains unresolved. The Brier score, the mean squared difference between the yes probability and binary label, rises from 0.0681 to 0.1035.

The six-by-six matrix of mean logit-margin changes has clinical directions as rows and questions as columns. Its first singular component accounts for 90.51% of the sum of squared matrix entries. Subtracting row and column means and restoring the grand mean leaves an interaction component with 7.91% of that same raw matrix energy. These are distinct decompositions with a common denominator, not complementary percentages. They summarize response geometry rather than identifying a shared causal channel. The answer-encoding experiment below tests another feature of that response structure: its behavior when the finding-to-letter mapping is exchanged.

## H ANSWER-ENCODING SENSITIVITY

### H.1 PAIRED DESIGN AND INDEPENDENT CAPABILITY CALIBRATION

The crossover selects the first 200 patients in the accepted causal-ownership matrix cohort’s registered order for interventions and the remaining 200 for capability calibration. Each patient contributes one image. These two sets are patient-disjoint from each other but both reuse the original ownership cohort. The six clinical questions cross three prompt packages, each question–prompt pair having its own unsteered baseline and 27 directions at each of  $\alpha = \pm 0.25$ : six clinical directions, 20 seed-0

random directions, and a coordinate- permutation sham. The 198,000 intervention outcomes and 2,400 clean calibration outcomes took 6,402 seconds on one A100.

Figure A8 displays the exact Mass prompts. The other five questions use the corresponding finding phrases from the original clinical family. Coded prompts exchange the answer mapping while retaining their wording; comparison with standard yes/no changes the question wording and response instructions as well as the answer symbols.

Is there a lung mass in this chest radiograph? Answer yes or no.

(a) Standard yes/no

Does this chest radiograph show a lung mass? Answer A if the finding is present and B if it is absent. Reply with A or B only.

(b) A-present

Does this chest radiograph show a lung mass? Answer B if the finding is present and A if it is absent. Reply with A or B only.

(c) B-present

Figure A8: **The crossover distinguishes answer mapping within coded prompts from the full prompt-package comparison.** The two coded conditions exchange presence and absence assignments for A and B. The standard condition retains the original question.

At the first answer position, the raw coded margin is the maximum A-token logit minus the maximum B-token logit among valid single-token leading-space variants. A-present uses that margin as the finding-present score; B-present negates it. Capability calibration uses these clinically oriented scores against image labels. Both mappings must have a one-sided 95% AUROC lower bound above 0.5 and at least ten patients in each label class. The lower bounds use 2,000 patient-bootstrap draws, with seed 20260908. Table A13 reports the resulting eligibility.

Table A13: Independent answer-encoding capability calibration.

| Question     | Positive | A-present<br>AUROC | B-present<br>AUROC | A-present<br>lower bound | B-present<br>lower bound | Eligibility              |
|--------------|----------|--------------------|--------------------|--------------------------|--------------------------|--------------------------|
| Effusion     | 11       | 0.5125             | 0.5111             | 0.3872                   | 0.3967                   | Below baseline criterion |
| Atelectasis  | 15       | 0.6586             | 0.6441             | 0.5448                   | 0.5372                   | Eligible                 |
| Pneumothorax | 13       | 0.6508             | 0.6524             | 0.5088                   | 0.5113                   | Eligible                 |
| Cardiomegaly | 8        | 0.5661             | 0.5811             | 0.4677                   | 0.4757                   | Insufficient labels      |
| Mass         | 15       | 0.6533             | 0.6542             | 0.5198                   | 0.5231                   | Eligible                 |
| Nodule       | 21       | 0.6347             | 0.6327             | 0.5210                   | 0.5196                   | Eligible                 |

Atelectasis, Pneumothorax, Mass, and Nodule enter the primary endpoint. Effusion and Cardiomegaly remain descriptive. Clean finding-present A/B rankings are highly correlated: independent calibration Spearman correlations range from 0.960770 to 0.988116, while the intervention-cohort range is 0.9594–0.9804. Calibration mean raw A-minus-B margins have opposite signs across mappings for every question; their absolute magnitudes differ by 0.0050–0.1200. Thus similar patient rankings coexist with baseline score asymmetry.

## H.2 MAPPING-EVEN AND MAPPING-ODD RESPONSE ENERGY

For patient  $i$ , direction  $d$ , and question  $q$ , let  $x_{i,d,q}^A$  and  $x_{i,d,q}^B$  denote the changes in raw A-minus-B margin from each prompt’s own baseline under A-present and B-present, respectively, at  $\alpha = +0.25$ . Define the mapping-even component  $e_{i,d,q}$  and mapping-odd component  $o_{i,d,q}$  by

$$e_{i,d,q} = \frac{1}{2}(x_{i,d,q}^A + x_{i,d,q}^B), \quad o_{i,d,q} = \frac{1}{2}(x_{i,d,q}^A - x_{i,d,q}^B). \quad (5)$$

The even component retains raw letter preference under the exchange; the odd component reverses raw sign as the clinical assignment reverses. These are coordinate decompositions, not mechanism labels.

For a sequence of  $n = 200$  patient effects  $z = (z_1, \dots, z_n)$ , let  $U(z)$  estimate its squared population mean. Let  $Q$  be the four eligible questions and  $\mathcal{D}$  the six clinical directions. The registered energy contrast  $T$  is

$$U(z) = \frac{(\sum_i z_i)^2 - \sum_i z_i^2}{n(n-1)}, \quad T = \frac{1}{6} \sum_{d \in \mathcal{D}} \sum_{q \in Q} [U(e_{\cdot, d, q}) - U(o_{\cdot, d, q})]. \quad (6)$$

The cross-patient products remove the finite-sample variance contribution to a squared sample mean. The endpoint sums questions and averages clinical directions, in squared-logit units. Joint patient resampling preserves pairing across mappings, directions, and questions; 2,000 draws use seed 20260907. Each random direction uses the same eligible-question sum. For sham, the question-specific permuted normal supplies each question’s effect. The registered even-beyond-controls condition requires a positive 95% lower endpoint for  $T$ , a point estimate above every random contrast, and a point estimate above the absolute sham contrast.

Table A14 compares the primary estimate with its controls and the all-question descriptive endpoint. Negative values favor mapping-odd squared mean response in the aggregate.

Table A14: Mapping-even versus mapping-odd response energy.

| Comparison             | Questions     | Estimate           | 95% interval       |
|------------------------|---------------|--------------------|--------------------|
| Clinical mean          | Four eligible | -2.1786            | [-2.4061, -1.9810] |
| Random-direction range | Four eligible | [-5.4773, -0.1769] | —                  |
| Sham                   | Four eligible | -2.3848            | [-2.5371, -2.2460] |
| Clinical mean          | All six       | -2.9329            | [-3.2374, -2.6636] |

The primary interval is entirely negative, and all 20 random-direction point estimates are negative as well. The registered even-beyond-controls condition is therefore unmet. Mapping-odd dominance is a property shared with generic perturbations in this experiment, so the energy contrast does not by itself establish clinical specificity or encoding invariance. The random row gives the observed range over directions, not an uncertainty interval; the other intervals are patient-bootstrap percentile summaries.

### H.3 PROMPT-CONDITIONED CLINICAL COMPETITION

Table A15 reports Mass in finding-present logit coordinates, using yes-minus-no for standard prompts, A-minus-B for A-present, and B-minus-A for B-present. Each clinical margin subtracts the maximum alternative among five directions within that same prompt; the maximum is recomputed in each patient-bootstrap draw. The matched effects under A-present and B-present are 1.6125 and 1.7469, exceeding their random maxima of 1.1650 and 1.2581 and absolute sham effects of 0.7550 and 1.1475. Their positive clinical margins contrast with the standard value of -0.7181 on the same 200 patients. That standard-prompt value is recomputed on the intervention subset; the full 400-patient diagnostic value is -0.7134 (Table A12).

Table A15: Mass clinical competition across prompt conditions.

| Prompt          | Clinical margin | 95% CI             | Mass effect | Random max | Sham   |
|-----------------|-----------------|--------------------|-------------|------------|--------|
| Standard yes/no | -0.7181         | [-0.7956, -0.6412] | 2.8625      | 3.0412     | 0.5869 |
| A-present       | 0.2825          | [0.2269, 0.3406]   | 1.6125      | 1.1650     | 0.7550 |
| B-present       | 0.2900          | [0.2300, 0.3506]   | 1.7469      | 1.2581     | 1.1475 |

This is a descriptive discovery selected from six questions, with pointwise intervals and 20 random directions. Standard-to-coded contrasts concern the whole prompt package because wording and response instructions also change. The independent confirmation below separates these prompt factors on fresh patients. The original 400-patient simultaneous ownership verdict concerns its original yes/no prompts.

## I INDEPENDENT MASS PROMPT CONFIRMATION

### I.1 COHORTS, PROMPT FACTORS, AND CALIBRATION

The prospective confirmation fixes Qwen2.5-VL-7B, the consumed `model.visual.blocks.31`,  $336 \times 336$  images, the six train-fitted clinical normals, and relative-token dose  $\alpha = +0.25$ . Its 200 patients are absent from the initial intervention, Effusion specificity, causal-ownership, and Consolidation closure cohorts. Selection uses only patient, row, and split metadata: sorted eligible patient IDs are permuted with seed 20260909, the first 200 are retained, and the same generator selects one sorted row per patient. The crossover’s 200 calibration patients are reused in their stored order and remain disjoint from confirmation. Calibration contains 15 positive and 185 negative patients; confirmation contains 12 positive and 188 negative patients.

Figure A9 shows the prompt factors. Each of the two question wordings is concatenated with each of the three explicit instructions, yielding six cells. A seventh anchor reproduces the original yes/no prompt. Thus the two *show* coded cells reproduce the discovery prompts exactly, while the factorial comparisons vary one factor at a time.

Does this chest radiograph show a lung mass?  
Is there a lung mass in this chest radiograph?

(a) Question wordings

Answer yes if the finding is present and no if it is absent. Reply with yes or no only.  
Answer A if the finding is present and B if it is absent. Reply with A or B only.  
Answer B if the finding is present and A if it is absent. Reply with A or B only.

(b) Explicit response instructions

Is there a lung mass in this chest radiograph? Answer yes or no.

(c) Original-prompt anchor

Figure A9: **The Mass confirmation crosses question wording with response instructions.** Each question in (a) combines with each instruction in (b); (c) supplies the seventh condition.

For each patient and cell, the finding-present logit margin is yes-minus-no, A-minus-B, or B-minus-A according to the response instruction. Scores use the maximum logit over accepted single-token spellings, including leading-space variants and the accepted yes/no case variants. Each intervention effect subtracts that cell’s own unsteered margin. The seven cells each receive a baseline, six clinical directions, 119 shared seed-0 isotropic random directions, and the Mass coordinate-permutation sham drawn after the random family. This yields 177,800 confirmation and 1,400 calibration outcomes. The complete run took 8,783 seconds on one A100.

A cell is eligible when calibration contains at least ten patients per class, at least 1,900 of 2,000 bootstrap draws have both classes, and the one-sided 95% AUROC lower bound exceeds 0.5. Calibration resampling uses seed 20260908. Table A16 reports the clean answer discrimination and Brier error for all seven cells.

Table A16: Mass capability calibration across prompt conditions.

| Condition              | AUROC  | One-sided 95% lower bound | Brier score |
|------------------------|--------|---------------------------|-------------|
| <i>show</i> /yes/no    | 0.6486 | 0.5105                    | 0.0703      |
| <i>show</i> /A-present | 0.6533 | 0.5198                    | 0.0894      |
| <i>show</i> /B-present | 0.6542 | 0.5231                    | 0.0939      |
| <i>is</i> /yes/no      | 0.6422 | 0.5043                    | 0.0720      |
| <i>is</i> /A-present   | 0.6420 | 0.5047                    | 0.1006      |
| <i>is</i> /B-present   | 0.6436 | 0.5051                    | 0.1041      |
| anchor                 | 0.6443 | 0.5088                    | 0.0716      |

All cells are eligible, with 2,000 valid draws each. Their AUROCs range from 0.6420 to 0.6542 and their lower bounds from 0.5043 to 0.5231. Eligibility is determined on calibration before inspecting



intervention outcomes; the confirmation cohort supplies a separate assessment of predictive behavior below.

## I.2 CLINICAL COMPETITION AND THE REGISTERED CONTROL FAMILY

The primary family comprises the four coded cells' five Mass-minus-other clinical contrasts, for 20 comparisons in total. Joint resampling of the 200 confirmation patients preserves pairing across cells and directions. A centered studentized max- $T$  bootstrap with 5,000 draws and seed 20260910 gives simultaneous one-sided 95% lower bounds. A coded cell qualifies when it is eligible, all five bounds are positive, and its positive Mass effect exceeds every same-cell random effect and the absolute sham effect. Cross-wording specificity requires all four coded cells to qualify; original-wording replication requires both *show* coded cells under the same joint family. The explicit yes/no cells and anchor are descriptive.

Table A17 reports the seven mean effects and clinical margins. A clinical margin subtracts the largest of the five competing clinical means within the cell. Its descriptive 95% percentile interval recomputes that maximum in each paired-patient bootstrap draw; it is distinct from the simultaneous pairwise bounds in Table 2.

Table A17: Mass clinical and generic-control competition.

| Condition              | Mass effect | Clinical margin | 95% interval       | Random max | Sham   |
|------------------------|-------------|-----------------|--------------------|------------|--------|
| <i>show</i> /yes/no    | 2.3800      | 0.2369          | [0.1362, 0.3394]   | 3.0531     | 0.1225 |
| <i>show</i> /A-present | 1.7019      | 0.2525          | [0.1869, 0.3175]   | 2.2100     | 0.2931 |
| <i>show</i> /B-present | 1.8188      | 0.2338          | [0.1675, 0.3006]   | 2.2169     | 0.2850 |
| <i>is</i> /yes/no      | 2.6600      | -0.4556         | [-0.5350, -0.3794] | 3.6031     | 0.6556 |
| <i>is</i> /A-present   | 1.9225      | -0.1087         | [-0.1606, -0.0575] | 2.4750     | 0.6075 |
| <i>is</i> /B-present   | 2.0019      | -0.2006         | [-0.2531, -0.1487] | 2.4944     | 0.5944 |
| anchor                 | 2.9987      | -0.7438         | [-0.8338, -0.6575] | 4.1312     | 0.9856 |

Effusion is the strongest clinical competitor in every cell. The *show* A-present and B-present clinical margins are positive, with minimum simultaneous lower bounds 0.1675 and 0.1462, respectively. Their Mass effects of 1.7019 and 1.8188 remain below random maxima 2.2100 and 2.2169. Two of the 119 random effects exceed Mass in each cell, giving reference ranks of 3/120; the criterion requires exceeding all 119. Under *is*, the two coded clinical margins are negative. Consequently, both registered specificity conjunctions are unmet. Clinical competition reproduces under the discovery wording, while the larger random family limits the supported specificity claim.

## I.3 PAIRED WORDING AND INSTRUCTION CONTRASTS

Table A18 isolates the observed changes in clinical competition. Each row subtracts two margins on the same patients, with the clinical maximum recomputed within each cell and resample. These eight paired percentile intervals are descriptive and do not change the two registered conjunctions.

Table A18: Paired changes in Mass clinical competition.

| Clinical-margin contrast                      | Difference | 95% interval       |
|-----------------------------------------------|------------|--------------------|
| <i>show</i> /A-present – <i>show</i> /yes/no  | 0.0156     | [-0.0250, 0.0544]  |
| <i>show</i> /B-present – <i>show</i> /yes/no  | -0.0031    | [-0.0425, 0.0350]  |
| <i>is</i> /A-present – <i>is</i> /yes/no      | 0.3469     | [0.3150, 0.3788]   |
| <i>is</i> /B-present – <i>is</i> /yes/no      | 0.2550     | [0.2219, 0.2881]   |
| <i>is</i> /yes/no – <i>show</i> /yes/no       | -0.6925    | [-0.7369, -0.6494] |
| <i>is</i> /A-present – <i>show</i> /A-present | -0.3612    | [-0.3906, -0.3319] |
| <i>is</i> /B-present – <i>show</i> /B-present | -0.4344    | [-0.4644, -0.4056] |
| <i>is</i> /yes/no – anchor                    | 0.2881     | [0.2631, 0.3137]   |

Under *show*, the coded-minus-explicit-yes/no intervals span zero; they leave the instruction contrast unresolved rather than establishing equivalence. Under *is*, both coded instructions improve the margin relative to explicit yes/no, although their margins remain negative. Within every response instruction, changing *show* to *is* lowers the clinical margin. Explicit yes/no instructions also improve the *is*

margin relative to the original anchor. The joint pattern distinguishes wording-conditioned clinical competition from a claim that A/B symbols alone create specificity.

#### I.4 ANSWER DISCRIMINATION AND PROBABILITY ERROR

Table A19 compares baseline and Mass-steered answer AUROC on the 200 confirmation patients. Scores are clinically oriented answer margins, as in capability calibration; they are not fitted probe scores. The paired change intervals use the same 5,000 patient resamples.

Table A19: Mass steering and answer discrimination.

| Condition             | Baseline AUROC | Steered AUROC | Change  | 95% interval      |
|-----------------------|----------------|---------------|---------|-------------------|
| <i>show/yes/no</i>    | 0.4938         | 0.5288        | 0.0350  | [−0.1505, 0.2129] |
| <i>show/A-present</i> | 0.5055         | 0.5461        | 0.0406  | [−0.1347, 0.2136] |
| <i>show/B-present</i> | 0.4962         | 0.5468        | 0.0505  | [−0.1307, 0.2223] |
| <i>is/yes/no</i>      | 0.4942         | 0.4989        | 0.0047  | [−0.1728, 0.1735] |
| <i>is/A-present</i>   | 0.4951         | 0.5202        | 0.0250  | [−0.1467, 0.1911] |
| <i>is/B-present</i>   | 0.4940         | 0.5441        | 0.0501  | [−0.1451, 0.2263] |
| anchor                | 0.4956         | 0.4807        | −0.0148 | [−0.1802, 0.1435] |

Baseline AUROCs lie near 0.5 in this cohort, and all seven paired AUROC-change intervals span zero. With only 12 positive confirmation patients, the intervals are broad. The calibration-based eligibility decision and this imprecise confirmation-cohort discrimination estimate answer different questions.

Table A20 evaluates probability error, using the sigmoid of each finding-present logit margin. The Brier score is the mean squared difference between that probability and the binary disease label; positive changes indicate larger error.

Table A20: Mass steering and probability error.

| Condition             | Baseline Brier | Steered Brier | Change | 95% interval     |
|-----------------------|----------------|---------------|--------|------------------|
| <i>show/yes/no</i>    | 0.0601         | 0.1530        | 0.0929 | [0.0691, 0.1163] |
| <i>show/A-present</i> | 0.0858         | 0.2949        | 0.2091 | [0.1829, 0.2338] |
| <i>show/B-present</i> | 0.0943         | 0.3131        | 0.2188 | [0.1911, 0.2445] |
| <i>is/yes/no</i>      | 0.0644         | 0.2238        | 0.1594 | [0.1305, 0.1871] |
| <i>is/A-present</i>   | 0.0985         | 0.3866        | 0.2881 | [0.2589, 0.3153] |
| <i>is/B-present</i>   | 0.1067         | 0.4015        | 0.2948 | [0.2644, 0.3227] |
| anchor                | 0.0630         | 0.2189        | 0.1559 | [0.1269, 0.1836] |

Mass steering raises finding-present probabilities while increasing Brier error in every cell, with all seven change intervals above zero. Thus a positive clinical competition margin under *show* coexists with worse probability predictions and unresolved discrimination change. These measurements concern one finding, model, visual locus, and dose; the wording contrasts characterize that intervention interface.

## J PAIRED BEHAVIORAL OPPORTUNITY ACROSS MODEL CONFIGURATIONS

### J.1 A SHARED, METADATA-MATCHED PATIENT COHORT

The paired-opportunity study evaluates 100 fresh test patients, each with one Consolidation-positive and one Consolidation-negative study. Patients are absent from the entire original probe manifest and all preceding intervention cohorts, including the earlier 50-pair closure study. Within each patient, the two studies match on view, sex, integer age, and all 13 other source disease labels. The fixed selection retains complete source records and patient/image order. Matching defines the paired population; unrecorded longitudinal differences can remain.

Figure A10 gives the common clinical prompt. Qwen is evaluated first, followed by a registered LLaVA comparison on exactly the same 200 images. The model-specific processor and chat template wrap that prompt. Qwen retains its accepted image processing; LLaVA uses its snapshot processor’s RGB conversion, 336-pixel shortest-edge resize,  $336 \times 336$  center crop, CLIP normalization, and default visual feature selection.

Is there consolidation in this chest radiograph? Answer yes or no.

Figure A10: **Both model configurations evaluate the same paired-input question.** The clinical prompt is fixed across the positive and negative studies and across models.

Each model supplies its first-answer yes-minus-no margin, taking the maximum over accepted singleton case and leading-space variants for each answer. The paired gap subtracts the negative study’s margin from the positive study’s margin in that model’s own coordinates. The sigmoid of each image’s margin gives its yes probability; the probability gap subtracts the two probabilities. This is a clean-response measurement, with no representation intervention. The Qwen and LLaVA runs each contain 200 image outcomes and take 31 and 38 seconds, respectively, on one A100.

### J.2 MODEL-SPECIFIC OPPORTUNITY AND THE DIRECT PAIRED COMPARISON

The registered opportunity endpoint is the patient-weighted mean margin gap within each model. Its one-sided 95% lower bound is the fifth percentile of 5,000 patient-bootstrap estimates. A strictly positive bound is required for positive opportunity. Both analyses use the identical stored resampling indices, generated with seed 20260912, and retain all 100 patients irrespective of the sign of their gap. Table A21 presents these model-specific endpoints and their descriptive two-sided percentile intervals.

Table A21: Uncertainty in paired Consolidation response.

| Model         | Mean margin gap | One-sided 95% lower bound | Descriptive 95% interval |
|---------------|-----------------|---------------------------|--------------------------|
| Qwen2.5-VL-7B | -0.1175         | -0.3987                   | [-0.4500, 0.2088]        |
| LLaVA-1.5-7B  | -0.0712         | -0.1338                   | [-0.1463, 0.0012]        |

Both lower bounds are negative, so neither configuration meets the opportunity criterion. The two-sided intervals include positive and negative mean responses: positive opportunity remains unresolved in this metadata-defined population. The model-specific margin scales are not subtracted from one another to infer a model difference.

For that direct comparison, within-patient concordance assigns one to a positive gap, zero to a negative gap, and one half to a tie. A tie is exact equality of the positive and negative margins under the registered scoring configured for BF16. A model’s concordance is the mean of these patient scores. The cross-model endpoint first subtracts Qwen’s score from LLaVA’s for each patient, then averages over the shared cohort. Its descriptive 95% interval reuses the same 5,000 patient-bootstrap draws. Table A22 shows the gap-sign counts and the direct contrast.

Table A22: Within-patient concordance and its direct model contrast.

| Model or contrast | Positive | Negative | Tied | Concordance | 95% interval    |
|-------------------|----------|----------|------|-------------|-----------------|
| Qwen2.5-VL-7B     | 54       | 45       | 1    | 0.545       | —               |
| LLaVA-1.5-7B      | 36       | 45       | 19   | 0.455       | —               |
| LLaVA – Qwen      | —        | —        | —    | −0.090      | [−0.210, 0.025] |

The direct interval spans zero, leaving the concordance difference unresolved. The sign counts account for every selected patient, and ties receive the same half-credit treatment in both models. The direct contrast uses the paired patient scores, rather than comparing the significance of the separate model-specific opportunity tests.

### J.3 PROBABILITY GAPS AND SOURCE-LABEL STRATA

Table A23 reports complementary descriptive summaries. The cohort contains 52 pairs whose negative image is labelled No Finding and 48 whose negative image has other source findings. Stratum means use the within-model logit gap; the full-cohort probability gap uses the difference of sigmoid scores defined above.

Table A23: Paired response across source-label strata.

| Model         | Full-cohort probability gap | No Finding margin gap ( $n = 52$ ) | Other findings margin gap ( $n = 48$ ) |
|---------------|-----------------------------|------------------------------------|----------------------------------------|
| Qwen2.5-VL-7B | −0.01743                    | −0.21154                           | −0.01563                               |
| LLaVA-1.5-7B  | −0.01508                    | −0.15385                           | 0.01823                                |

Mean probability gaps are negative in both configurations. The stratum summaries describe the selected population and leave the registered full-cohort opportunity decisions unchanged. Taken together, the mean-gap uncertainty and the direct concordance comparison identify the behavioral opportunity that remains unresolved before interpreting component-based recovery. Because this cohort differs from the original 50-pair closure cohort, it contextualizes that study rather than identifying the cause of its closure result.

## K FROZEN READERS AND CLEAN ANSWER DISCRIMINATION IN LLaVA

### K.1 PROSPECTIVE VALIDATION ALLOCATION AND QUALIFICATION

The validation screen asks which fixed clinical directions combine controlled readability with an informative clean answer margin before testing their write efficacy. It reuses the accepted LLaVA-1.5-7B snapshot, native image processor, final visual block, and six train-fitted clinical directions. The clinical questions are the six prompts in Figure A1. The scoring interface takes the maximum first-answer logit among accepted singleton yes variants minus the maximum among no variants, retaining case and leading-space variants. Its sigmoid is a model score rather than a calibrated disease probability.

Allocation uses the original manifest’s validation split, excluding every patient represented in the sixteen implementation-preflight images. Remaining patient identifiers are sorted and permuted with seed 20260913. The first 700 patients form the reader and capability cohort, and the next 100 are reserved for conditional write evaluation. Each patient contributes its lexicographically smallest validation image identifier. This allocation precedes inspection of labels and responses. The two cohorts are patient-disjoint, as are the original training, validation, and test splits. All six questions use the same 700 calibration images.

Clinical reader scores use the stored projection, training centering and scaling, and logistic coefficients. Twenty recurring-type random-label controls are fitted on the original training split and shared across the six comparisons. No reader or control parameter is fitted on validation. Selectivity, defined in Section 3, subtracts mean control AUROC from clinical AUROC, with each control evaluated against its own type labels. The same 2,000 patient-bootstrap draws, seeded with 20260914, serve every clinical reader, every control, and clean answer capability.

Reader qualification requires a positive 2.5th-percentile selectivity bound. Clean capability requires an answer-margin AUROC fifth-percentile bound strictly above 0.5. Both require at least ten patients per label class and 1,900 valid draws. A question enters the write screen only when both criteria pass. These prospectively fixed thresholds define an exploratory validation policy; their percentile intervals are not selection-adjusted confirmatory intervals across the six questions.

### K.2 CONTROLLED READER AVAILABILITY

Table A24 presents all six frozen-reader comparisons. Every question meets the class-count requirement, and every reader comparison yields all 2,000 valid joint draws.

Table A24: Frozen clinical-reader qualification on validation patients.

| Question     | Positive | Negative | Selectivity | 95% interval      | Qualifies |
|--------------|----------|----------|-------------|-------------------|-----------|
| Effusion     | 34       | 666      | 0.1438      | [0.0658, 0.2163]  | Yes       |
| Atelectasis  | 43       | 657      | 0.0472      | [−0.0351, 0.1198] | No        |
| Pneumothorax | 12       | 688      | 0.0362      | [−0.1271, 0.1731] | No        |
| Cardiomegaly | 25       | 675      | 0.0889      | [0.0002, 0.1723]  | Yes       |
| Mass         | 39       | 661      | 0.0735      | [−0.0023, 0.1473] | No        |
| Nodule       | 48       | 652      | 0.0037      | [−0.0833, 0.0865] | No        |

Effusion and Cardiomegaly satisfy the registered reader screen. Effusion’s lower bound is 0.065803, whereas Cardiomegaly’s is 0.000164 before rounding. The latter sits close to the qualification threshold. The other four intervals span zero. Effusion’s larger pilot lower bound motivated its selection for independent answer and reader validation. That follow-up does not replicate its positive selectivity (Appendix L); the pilot selection remains exploratory.

### K.3 CLEAN CAPABILITY AND THE CONDITIONAL WRITE STAGE

Table A25 reports clean answer-margin AUROC on the same patients. Its two-sided 95% intervals are descriptive; the separate one-sided 95% lower bounds implement the capability rule. All six questions again have 2,000 valid draws. The run measures 4,200 clean image-question outcomes in 366 seconds on one A100.

Table A25: Clean answer discrimination across the six validation questions.

| Question     | AUROC  | Descriptive 95% interval | One-sided 95% lower bound | Qualifies |
|--------------|--------|--------------------------|---------------------------|-----------|
| Effusion     | 0.4548 | [0.3546, 0.5555]         | 0.3722                    | No        |
| Atelectasis  | 0.5275 | [0.4473, 0.6111]         | 0.4609                    | No        |
| Pneumothorax | 0.5122 | [0.3785, 0.6400]         | 0.4028                    | No        |
| Cardiomegaly | 0.5232 | [0.4034, 0.6375]         | 0.4240                    | No        |
| Mass         | 0.5518 | [0.4584, 0.6514]         | 0.4723                    | No        |
| Nodule       | 0.5536 | [0.4601, 0.6482]         | 0.4731                    | No        |

The largest capability lower bound is 0.473107, so no question qualifies jointly for reader availability and clean capability. The reserved 100-patient write cohort remains untouched. Its planned bidirectional clinical, random, and sham comparisons therefore contribute no efficacy estimate. This is the registered outcome of the conditional screen.

The pilot tables juxtapose controlled reader information and unresolved clean answer discrimination. These criteria test different quantities; their qualification pattern is not a direct statistical comparison. Appendix L reports the ensuing finite text calibration and independent image/reader test. Its nonreplicated selectivity limits the pilot pattern to this cohort rather than establishing a reproducible separation between measurement stages.

## L TEXT CALIBRATION AND INDEPENDENT IMAGE/READER VALIDATION

### L.1 FINITE CALIBRATION OF THE ANSWER INTERFACE

The pilot motivates two distinct checks: whether answer instructions preserve known textual meaning, and whether the selected interface and frozen reader generalize to independent images. The text check uses 16 constructed descriptions: two target objects (red square and blue circle), four positive/negative patterns, and both truth values. Every prompt begins with “The target finding is a {object}.” The object placeholder is replaced literally throughout. Figure A11 shows all description patterns and response instructions. Their order and the routing rule are fixed before observing outcomes.

The scene contains a {object}. / The scene contains no {object}.  
A {object} is present. / A {object} is absent.  
It is false that the {object} is absent. / It is false that the {object} is present.  
The list is complete. Listed objects: {object}, green triangle. /  
The list is complete. Listed objects: green triangle.

(a) Positive / negative description pairs

Is the finding present?  
Answer yes or no.

(b) Yes/no question and instruction

Answer A if the finding is present and B if it is absent. Reply with A or B only.

(c) Incumbent A-present instruction

Choose the correct option.  
A. The finding is present.  
B. The finding is absent.  
Reply with A or B only.

(d) Candidate A-present instruction

Figure A11: **Constructed descriptions and answer instructions for text calibration.** Each description follows the target preface and precedes the question in (b). The A/B conditions replace only the answer instruction; B-present swaps the letter assignments in (c), or the option meanings in (d).

All five conditions share the question “Is the finding present?” The yes/no margin takes the maximum yes singleton logit minus the maximum no singleton logit; a secondary margin replaces each maximum with log-sum-exp over the same variants. The A/B margin uses singleton token IDs 319 and 350 and is oriented toward presence. Each margin must be strictly positive for presence and strictly negative for absence; zero is an error. A yes/no route requires both aggregations to be correct on all 16 descriptions and at least one candidate A/B condition to make an error. This finite constructed-suite rule supplies no population or clinical inference. The 80 text outcomes take 15 dispatcher seconds on one A100.

Table A26 reports every condition. Its minimum correctly oriented margin multiplies a presence-oriented score by +1 for a positive description and by −1 for a negative description. Yes/no passes both aggregations (the secondary minimum is 0.12500056). Each candidate A/B condition fails the two negative complete-list descriptions, one per object. The registered rule therefore selects yes/no.

Table A26: Answer-sign accuracy on constructed descriptions.

| Condition           | Primary errors / 16 | Minimum oriented margin | LSE errors / 16 |
|---------------------|---------------------|-------------------------|-----------------|
| Yes/no              | 0                   | 0.125                   | 0               |
| Incumbent A-present | 7                   | −3.500                  | —               |
| Incumbent B-present | 8                   | −1.000                  | —               |
| Candidate A-present | 2                   | −2.000                  | —               |
| Candidate B-present | 2                   | −1.500                  | —               |

Table A27 gives the positive-minus-negative primary-margin change for each description pair, in Figure A11(a) order. All changes are positive, including conditions with sign errors. Sensitivity to a textual truth change therefore does not by itself establish correct individual answer semantics. These results select an interface for image testing, not a clinically capable model.

Table A27: Paired text-margin changes across description patterns.

| Condition           | Object      | Contains | Present/absent | Negation | Complete list |
|---------------------|-------------|----------|----------------|----------|---------------|
| Yes/no              | Red square  | 5.875    | 5.625          | 5.250    | 1.125         |
|                     | Blue circle | 5.750    | 5.500          | 6.000    | 1.125         |
| Incumbent A-present | Red square  | 3.500    | 3.375          | 3.375    | 0.125         |
|                     | Blue circle | 3.375    | 3.125          | 2.875    | 0.250         |
| Incumbent B-present | Red square  | 0.750    | 0.250          | 0.875    | 0.250         |
|                     | Blue circle | 1.125    | 0.500          | 1.125    | 0.250         |
| Candidate A-present | Red square  | 5.500    | 5.875          | 5.750    | 0.250         |
|                     | Blue circle | 5.250    | 6.250          | 5.250    | 0.375         |
| Candidate B-present | Red square  | 4.250    | 4.500          | 5.375    | 0.375         |
|                     | Blue circle | 4.125    | 5.125          | 5.750    | 0.375         |

## L.2 INDEPENDENT PAIRED IMAGES AND FROZEN MEASUREMENTS

The image cohort contains validation patients absent from the entire original 26,229-row manifest. The lexicographically first image identifier represents each patient. String-sorted patient identifiers are permuted with NumPy seed 20260916: the first 700 supply index images and the next 700 supply donors, paired by position. All 1,400 patients are distinct and independent of the pilot cohort. Index labels contain 25 Effusion-positive and 675 negative cases; donor labels contain 22 positive and 678 negative cases. Selection is label-blind within this pool. The pilot instead samples within the original disease-positive/No Finding-sampled manifest, so the two cohorts do not share the same sampling frame.

Both wordings use the accepted LLaVA-1.5-7B snapshot, native 336-pixel processor and chat template, configured for BF16 with batch size 16. The primary score is the first-answer maximum-variant yes-minus-no logit margin, using singleton yes IDs 3869, 4874, 22483 and no IDs 694, 1939, 11698. Logits are retained in float32; the score’s sigmoid is not a calibrated disease probability. Secondary log-sum-exp scores reuse those same logits. The run measures 2,800 image outcomes and two fixed-prompt image-free priors in 260 dispatcher seconds on one A100.



Figure A12 gives the exact image prompts. Let  $w \in \{0, 1\}$  index the accepted and alternative wording, respectively. Let  $A_w$  be real-image score AUROC against index labels,  $D_w$  donor-image score AUROC against those same index labels, and  $G_w = A_w - D_w$  the image advantage. Donor-own-label AUROC is a separate descriptive diagnostic, not the donor term in  $G_w$ .

Is there a pleural effusion in this chest radiograph? Answer yes or no.

(a) Accepted wording

Does this chest radiograph show a pleural effusion? Answer yes or no.

(b) Alternative wording

Figure A12: **Fixed Effusion prompts for independent image validation.** Each prompt is evaluated with index images, donor images, and once without an image. The response instruction is shared across wordings.

### L.3 IMAGE OPPORTUNITY AND WORDING UNDER THE PRIMARY FAMILY

Table A28 presents descriptive 95% percentile intervals. The resampling unit is a complete index-donor pair, shared across both wordings. All 10,000 paired bootstrap draws (seed 20260917) are valid; AUROC assigns half credit to ties. Each individual interval describes its own quantity and does not implement the registered joint decision.

Table A28: Real-image and donor-image answer discrimination.

| Quantity                         | Accepted wording  | Alternative wording |
|----------------------------------|-------------------|---------------------|
| Real AUROC $A_w$                 | 0.5259            | 0.5089              |
| Descriptive 95% interval         | [0.4135, 0.6367]  | [0.3916, 0.6255]    |
| Donor AUROC $D_w$ (index labels) | 0.4072            | 0.3975              |
| Descriptive 95% interval         | [0.3085, 0.5107]  | [0.2868, 0.5113]    |
| Image advantage $G_w$            | 0.1187            | 0.1113              |
| Descriptive 95% interval         | [-0.0013, 0.2383] | [-0.0357, 0.2583]   |
| Donor-own-label AUROC            | 0.4542            | 0.5117              |

The registered primary family contains  $(A_0 - 0.5, G_0, G_1 - G_0)$ . For each common valid draw, we take the maximum absolute deviation of these three bootstrap estimates from their observed estimates. The 95th percentile of those maxima, using NumPy's linear quantile, supplies a common radius of 0.1304. Table A29 reports each point estimate plus or minus this radius. These are untruncated, unstudentized simultaneous intervals, not studentized max- $T$  intervals. The rule requires at least 9,500 draws with all four AUROCs and all three contrasts finite. Image opportunity requires strictly positive lower bounds for both  $A_0 - 0.5$  and  $G_0$ ; wording dependence requires the third interval to exclude zero.

Table A29: Simultaneous image-opportunity and wording contrasts.

| Contrast    | Estimate | Simultaneous 95% interval |
|-------------|----------|---------------------------|
| $A_0 - 0.5$ | 0.0259   | [-0.1045, 0.1563]         |
| $G_0$       | 0.1187   | [-0.0117, 0.2491]         |
| $G_1 - G_0$ | -0.0073  | [-0.1377, 0.1231]         |

Neither opportunity lower bound is positive. The cross-wording interval also spans zero, leaving wording dependence unresolved rather than establishing no effect or equivalence. The descriptive real-image AUROC difference  $A_1 - A_0$  is -0.0170 (95% interval [-0.0650, 0.0300]).

#### L.4 INDEPENDENT FROZEN-READER REPLICATION

The Effusion reader uses index images under the accepted wording. Its features average all 577 tokens at the consumed vision block `model.vision_tower.encoder.layers.22`, stored in float16. The projection, training mean and scale, clinical coefficients, and all 20 control models and type maps are reused without fitting. Controls predict fixed random binary labels assigned to recurring view–sex–age–decade types; each is evaluated against its own type labels, not against Effusion labels. Selectivity subtracts their mean AUROC from clinical AUROC, as defined in Section 3. It is not a direct estimate of deconfounded clinical information.

Table A30 compares the pilot and independent cohort. The frozen-reader analysis uses the same 10,000 bootstrap indices as the answer analysis but a separate validity mask requiring finite clinical and all 20 control AUROCs. All draws are valid. The registered replication criterion requires at least ten cases per class, 9,500 valid draws, and a positive 2.5th-percentile selectivity bound. That bound is negative, so the pilot’s positive selectivity does not replicate.

Table A30: Frozen-reader performance across the two cohorts.

| Quantity                     | Pilot            | Independent index cohort |
|------------------------------|------------------|--------------------------|
| Effusion-positive / negative | 34 / 666         | 25 / 675                 |
| Clinical AUROC               | 0.8196           | 0.6664                   |
| Mean control AUROC           | 0.6758           | 0.6775                   |
| Selectivity                  | 0.1438           | −0.0111                  |
| Selectivity 95% interval     | [0.0658, 0.2163] | [−0.1541, 0.1185]        |
| Positive lower bound         | Yes              | No                       |

At the point-estimate level, the selectivity difference primarily reflects lower clinical AUROC, not a higher control mean. The pilot clinical AUROC interval is [0.7424, 0.8920]; no between-cohort difference interval was registered. Different sampling frames, limited positive cases, and selection of Effusion from the pilot are unresolved explanations, not identified mechanisms. Reader results neither filter the answer grid nor alter its simultaneous decision.

Table A31 lists every frozen control in its stored order, making the reported mean auditable. Their task-dependent AUROCs describe recurring nuisance-type predictability and should not be read as 20 alternative clinical Effusion classifiers.

Table A31: Frozen nuisance-type control AUROCs on independent patients.

| Control | AUROC  | Control | AUROC  | Control | AUROC  | Control | AUROC  |
|---------|--------|---------|--------|---------|--------|---------|--------|
| 1       | 0.9000 | 6       | 0.7416 | 11      | 0.6806 | 16      | 0.7076 |
| 2       | 0.5961 | 7       | 0.5952 | 12      | 0.5961 | 17      | 0.5706 |
| 3       | 0.6346 | 8       | 0.7585 | 13      | 0.7022 | 18      | 0.6587 |
| 4       | 0.7142 | 9       | 0.6728 | 14      | 0.6036 | 19      | 0.7593 |
| 5       | 0.7397 | 10      | 0.6238 | 15      | 0.6194 | 20      | 0.6749 |

## L.5 DESCRIPTIVE AGGREGATION AND PRIOR DIAGNOSTICS

Table A32 recomputes discrimination from the same logits using log-sum-exp (LSE). The accepted-wording image advantage has a positive descriptive percentile interval, but this secondary endpoint does not replace the primary simultaneous family. Intervals for the paired difference from the primary maximum-variant advantage span zero under both wordings. The evidence does not isolate aggregation choice as the source of unresolved primary opportunity.

Table A32: Same-logit aggregation sensitivity.

| Quantity                       | Accepted wording  | Alternative wording |
|--------------------------------|-------------------|---------------------|
| LSE real-image AUROC           | 0.5263            | 0.5075              |
| LSE donor AUROC (index labels) | 0.3990            | 0.3944              |
| LSE image advantage            | 0.1273            | 0.1131              |
| Descriptive 95% interval       | [0.0019, 0.2504]  | [-0.0377, 0.2637]   |
| LSE minus primary advantage    | 0.0087            | 0.0017              |
| Descriptive 95% interval       | [-0.0053, 0.0220] | [-0.0137, 0.0166]   |

Table A33 reports margin means, Brier error of the sigmoid score, and total probability mass on the accepted answer tokens. Real and donor Brier errors both use index labels. The alternative wording shifts both image-source margins toward yes and has higher Brier error; its real-image AUROC and paired image advantage remain unresolved under their respective intervals. This pattern distinguishes a score/prior shift from established improvement in disease discrimination.

Table A33: Answer margins, predictive error, and token mass.

| Input       | Quantity               | Accepted wording | Alternative wording |
|-------------|------------------------|------------------|---------------------|
| Real image  | Mean primary margin    | 0.0488           | 0.7998              |
|             | Brier error            | 0.2728           | 0.4604              |
|             | Mean answer-token mass | 0.9972           | 0.9965              |
| Donor image | Mean primary margin    | 0.0380           | 0.7907              |
|             | Brier error            | 0.2726           | 0.4596              |
|             | Mean answer-token mass | 0.9972           | 0.9965              |
| No image    | Primary margin         | 0.8750           | 1.2500              |
|             | Answer-token mass      | 0.9932           | 0.9926              |
|             | Constant-score AUROC   | 0.5000           | 0.5000              |

Each image-free prompt produces one constant score, whose AUROC against index labels is 0.5 under half-credit ties. These are prior and token-mass diagnostics, not patient-varying no-image discrimination estimates. Subtracting a wording-specific constant from all its image scores leaves AUROC unchanged. High answer-token mass establishes concentration on the chosen verbalizers, not clinical correctness.

Taken together, finite text calibration selects a semantically consistent yes/no interface for the constructed suite. Independent images establish neither the registered opportunity conjunction nor wording dependence, and the frozen clinical reader does not reproduce its pilot selectivity. The pilot remains an observation within its sampling frame; it does not support a reproducible reader-answer separation or identify the mechanism behind the downstream write-control outcomes.